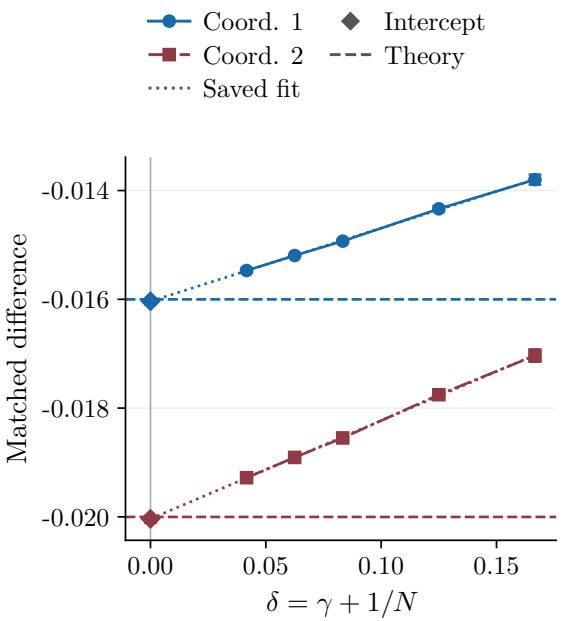
Figure 1 | Matched SCAFFOLD-FedAvg coefficients

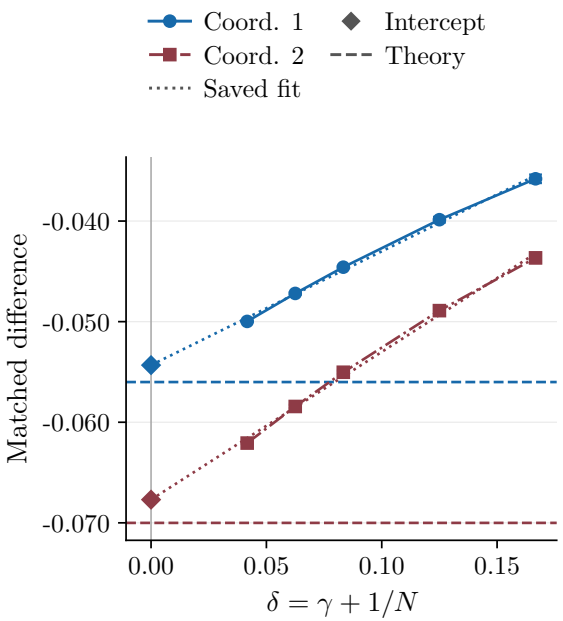
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H = 2

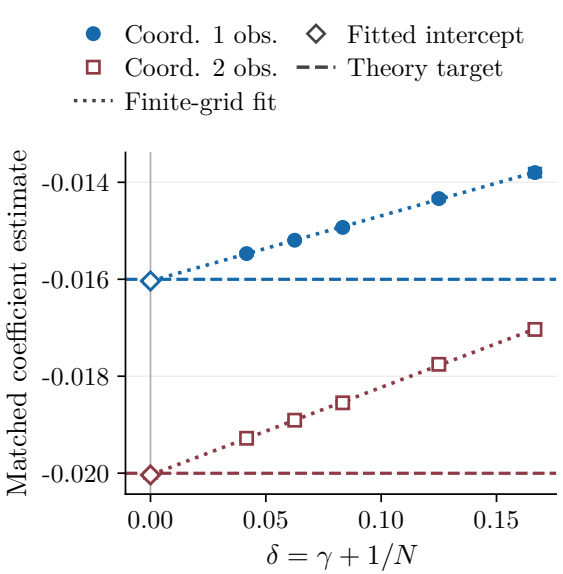


H = 4



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H = 2



H = 4

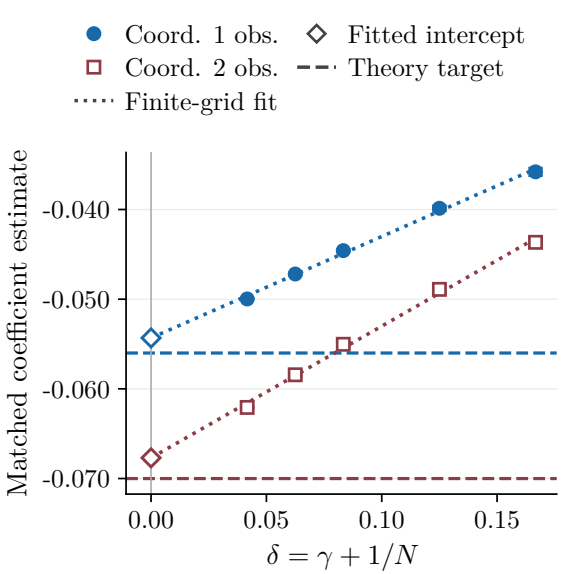
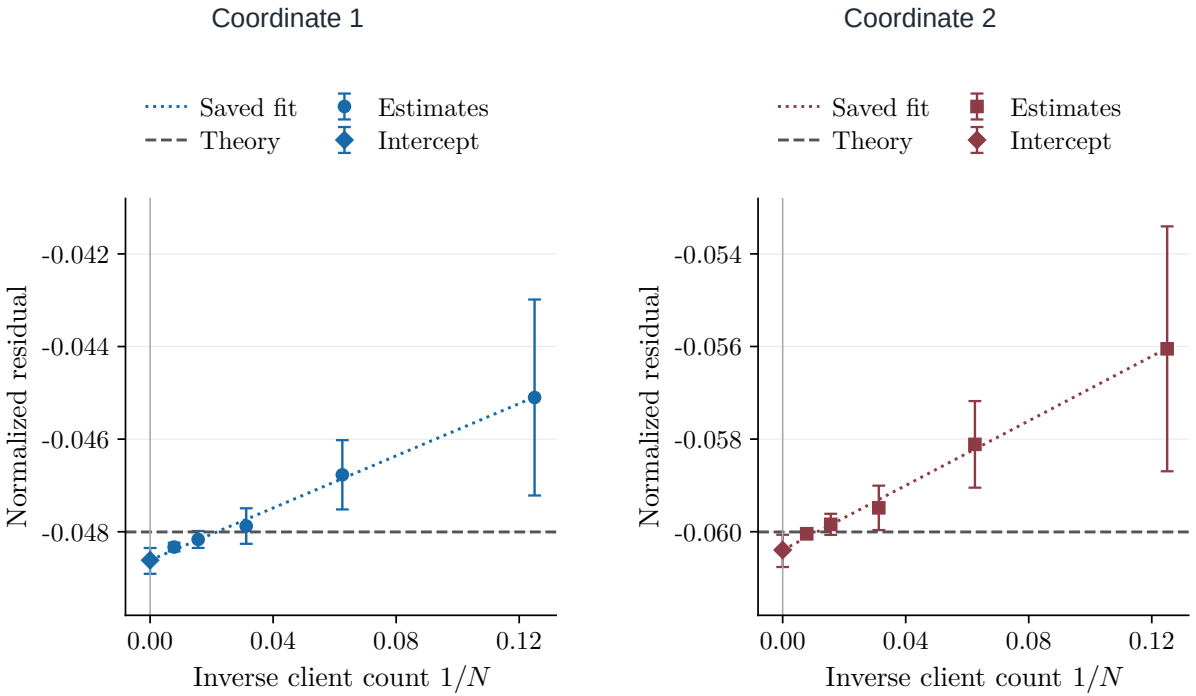


Figure 2 | Fixed-step client-count experiment

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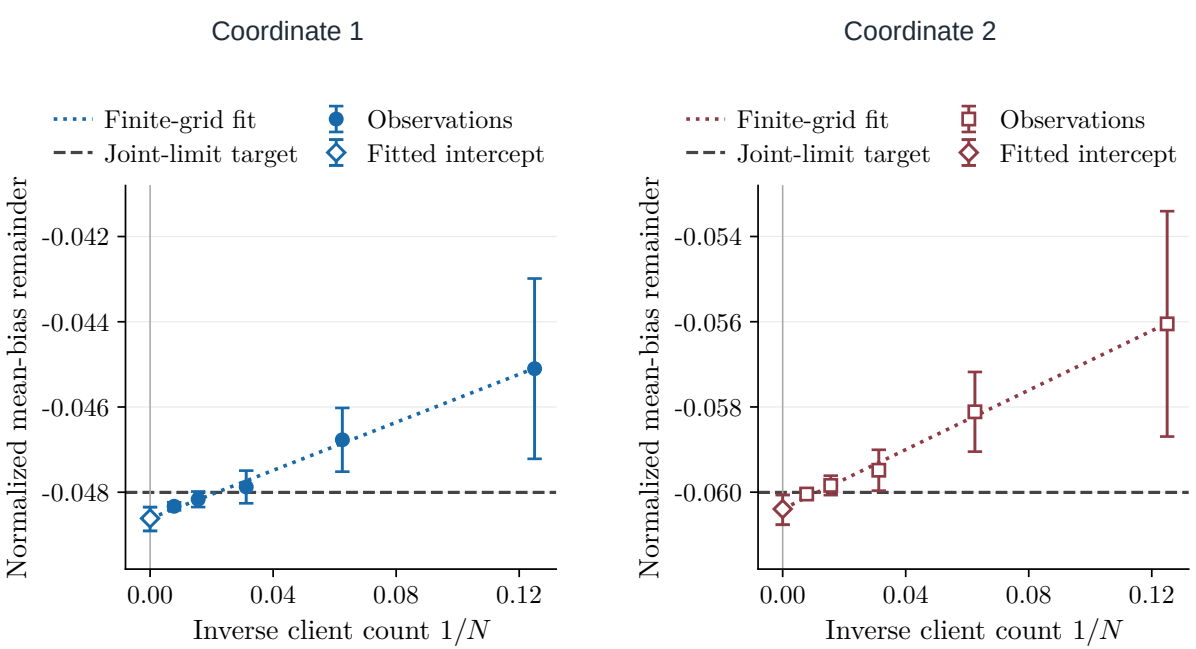
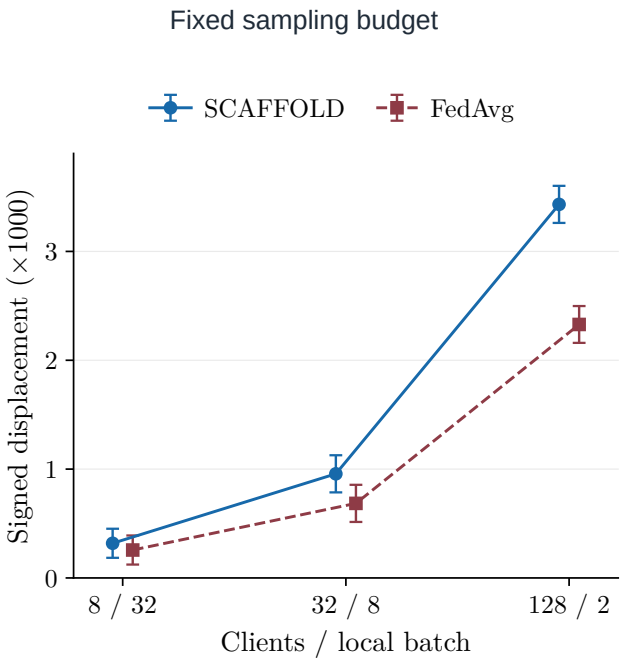


Figure 3 | Real-data allocation comparison

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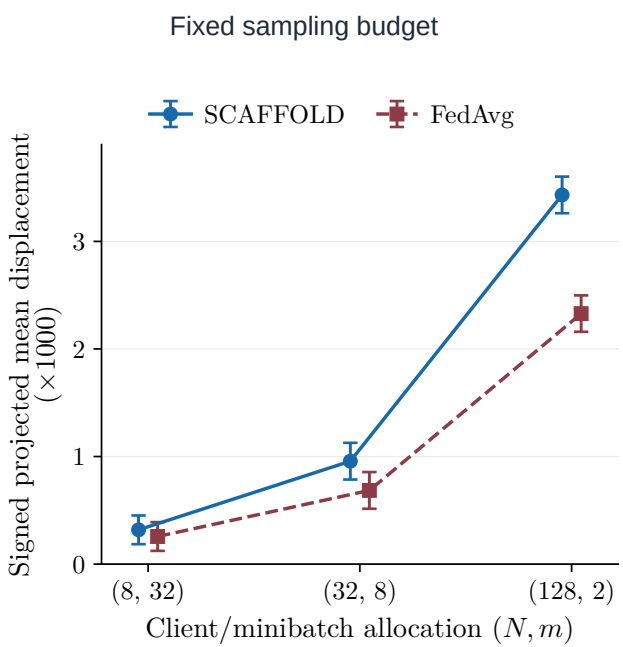
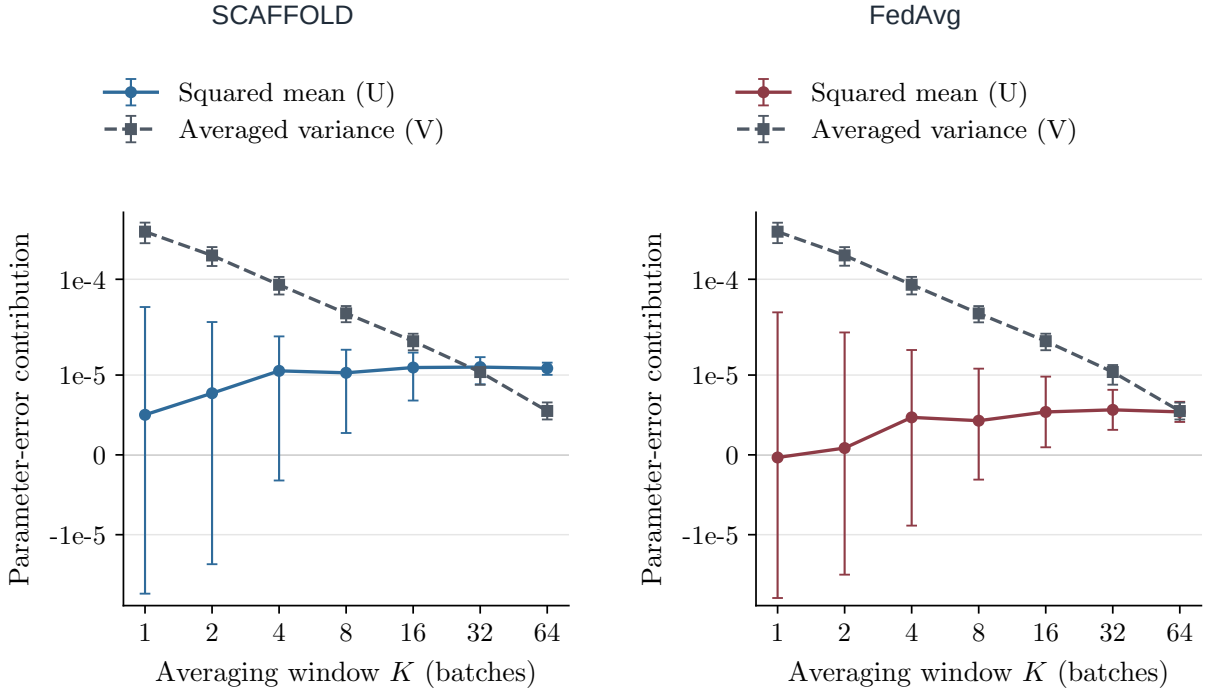


Figure 4 | Temporal model averaging

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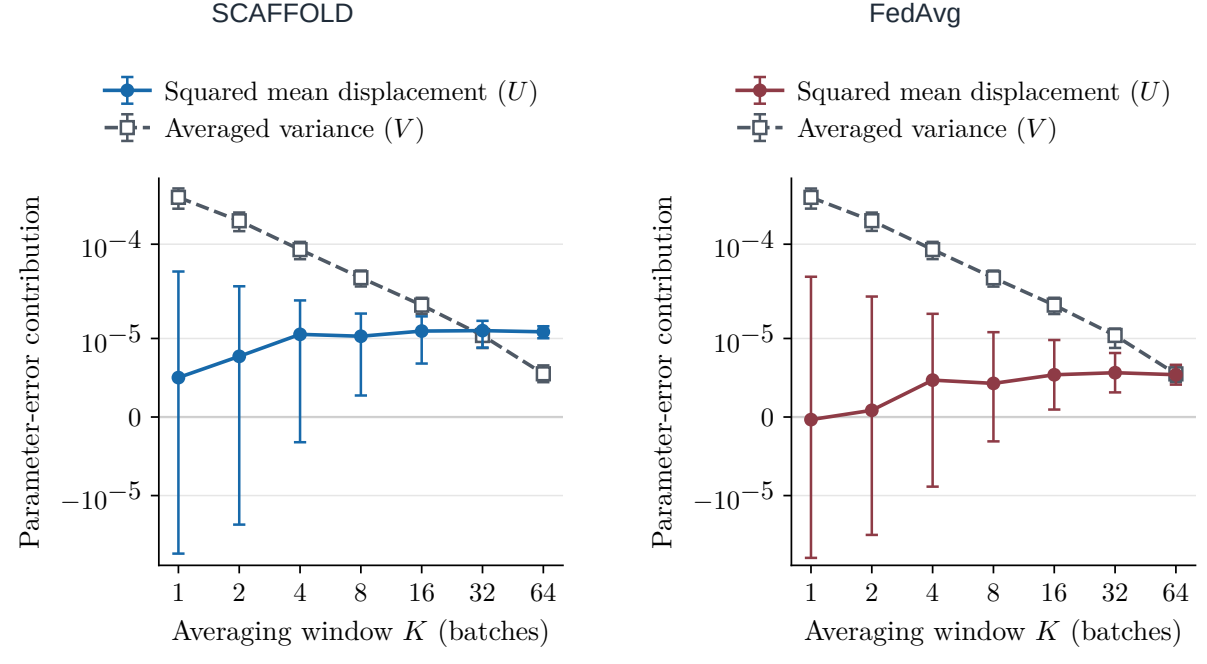
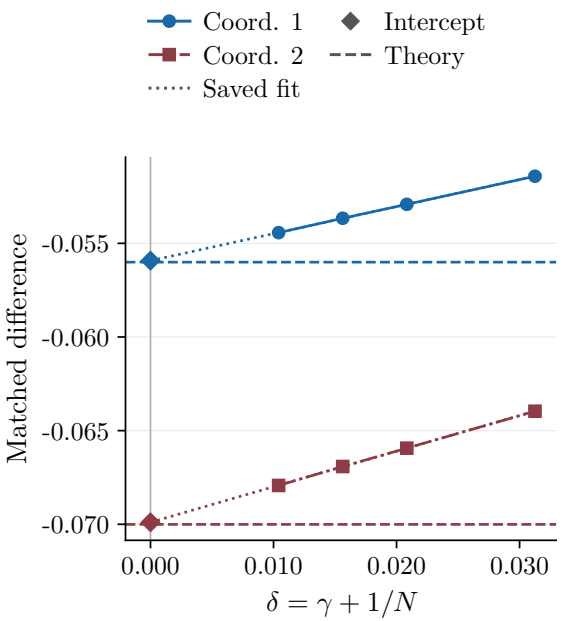


Figure 5 | H = 4 finite-grid diagnostics

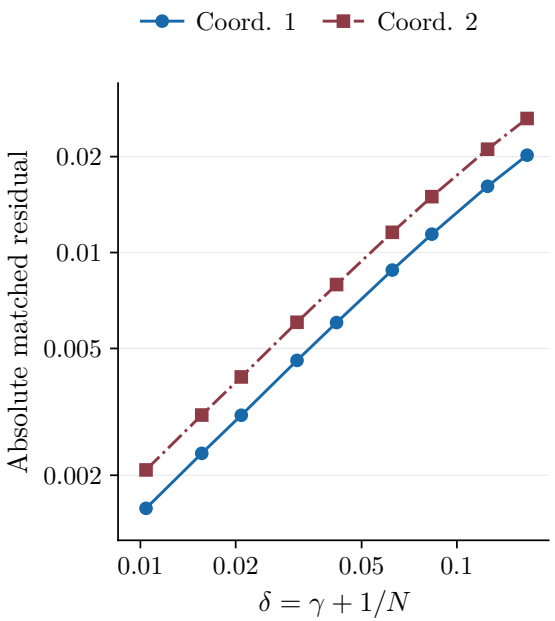
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Smaller-step grid

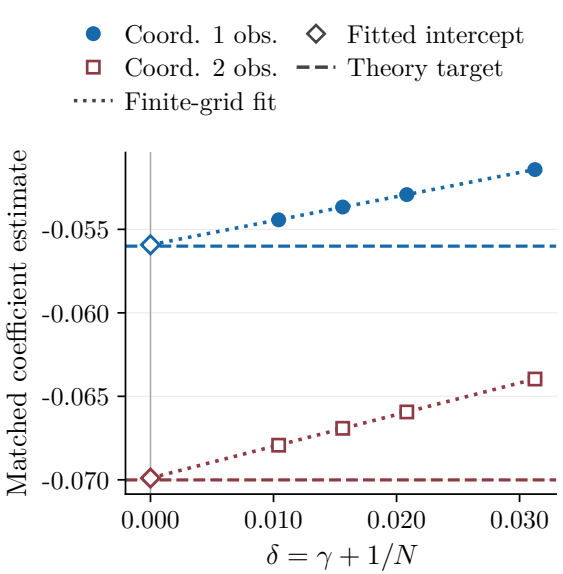


Residual diagnostic

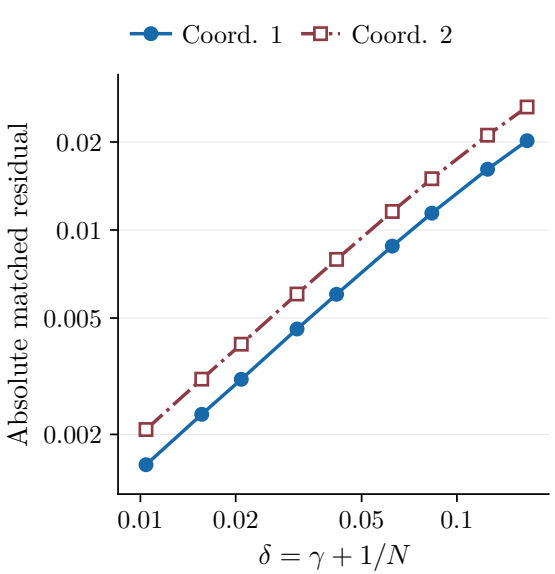


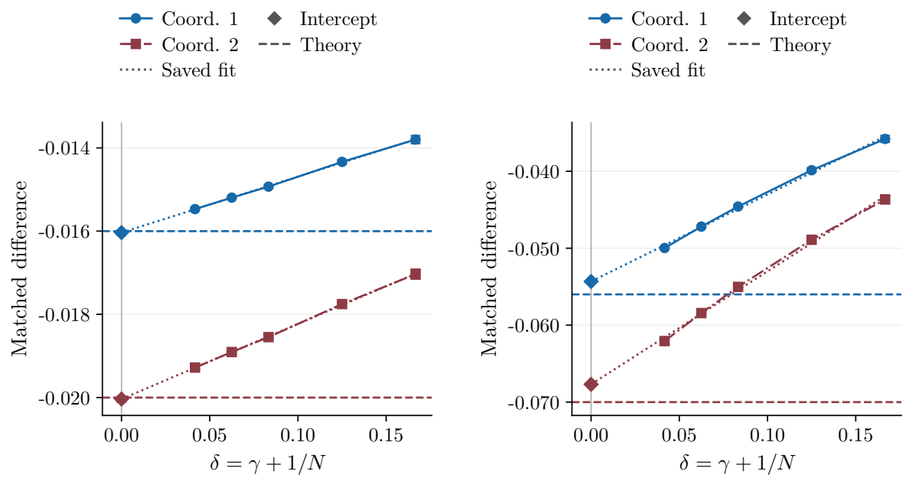
AFTER

Smaller-step grid



Residual diagnostic





(a) Two local steps

(b) Four local steps

Figure 1: Matched SCAFFOLD–FedAvg coefficient estimates for (a) $H = 2$ and (b) $H = 4$ on the primary joint grids. The ordinate is $\widehat{D}_j$ from Equation (34); dashed lines give the analytical targets and dotted curves give linear extrapolations. Pointwise bars use simultaneous Student-t intervals; intercept bars use simultaneous seed-block-bootstrap intervals.

6.2.5 Local-step dependence on an independent grid

Design and rationale. The theory assigns different roles to local training length: B_{11} is shared across H, whereas $B_{20}^{\mathrm{ctrl}}(H)$ changes explicitly with H. An independent implementation tests both predictions. Individual fits of $(N/\gamma)\widehat{b}$ estimate B_{11} freely. Matched fits on a Cartesian grid vary γ and $1/N$ independently, allowing a direct comparison with the predicted relation $B_{20}^{\mathrm{ctrl}}(4) = 3.5B_{20}^{\mathrm{ctrl}}(2)$.

Results and interpretation. The leading-coefficient intervals contain their targets for $H = 1, 2, 4$; both algorithms have offsets at $H = 8$. The adjusted SCAFFOLD $H = 8$ versus $H = 2$ leading-intercept contrasts exclude zero (Table 15). For the matched coefficient, Table 3 shows target inclusion at $H = 2, 4$ and target exclusion at $H = 8$. The adjusted $H = 4$ minus $3.5(H = 2)$ contrast contains zero in both coordinates. These independent estimates support the predicted local-step relation at $H = 2, 4$ and identify $H = 8$ as more sensitive to the fitted grid. Restricted-grid estimates and the separate joint-grid $H = 4$ discrepancy are documented in Appendix C.2.

6.2.6 Affine-shift equivalence

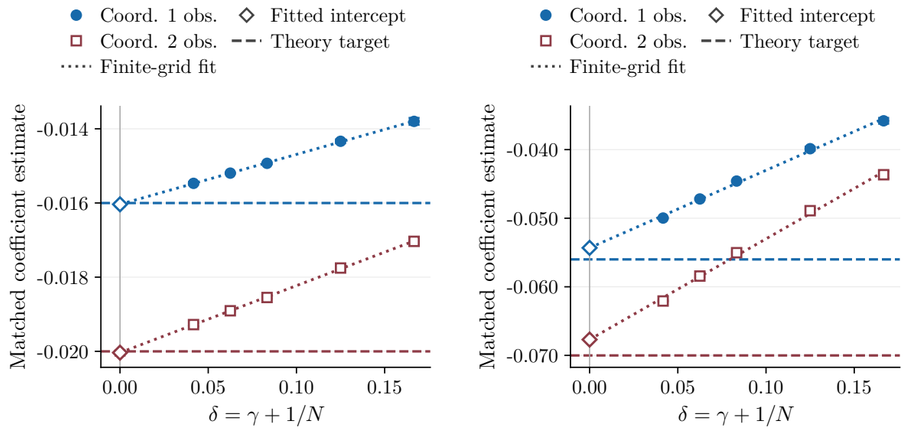
Design and rationale. Proposition 4.4 predicts exact equivalence after client-specific gradient offsets are absorbed into the controls. We test this identity by pairing affine-shift and homogeneous recursions with translated initial controls and identical noise, across several shift sizes, client counts, and local-step counts.

Results and interpretation. Across 24 configurations of 2,000 rounds, the maximum control-state residual is 1.64313×10^{-14}. The numerical recursions therefore reproduce the predicted

Table 2: Analytical coefficient vectors and fitted finite-grid intercepts. Individual rows target $B_{20}^{A}(H)$; matched-pair rows target $B_{20}^{\mathrm{ctrl}}(H)$. P and S denote the primary and smaller-step joint grids in Table 7. Values are rounded to seven decimal places; simultaneous intervals are in Table 11.

Study	H	Grid	Target vector	Estimated intercept vector
SCAFFOLD	2	P	$(-0.0480040, -0.0600050)$	$(-0.0476713, -0.0595811)$
SCAFFOLD	4	P	$(-0.1520127, -0.1900159)$	$(-0.1502614, -0.1874524)$
SCAFFOLD	4	S	$(-0.1520127, -0.1900159)$	$(-0.1514772, -0.1893273)$
FedAvg	2	P	$(-0.0320027, -0.0400033)$	$(-0.0320430, -0.0400539)$
FedAvg	4	P	$(-0.0960080, -0.1200100)$	$(-0.0959717, -0.1198059)$
FedAvg	4	S	$(-0.0960080, -0.1200100)$	$(-0.0959727, -0.1199590)$
Paired difference	2	P	$(-0.0160013, -0.0200017)$	$(-0.0160363, -0.0200366)$
Paired difference	4	P	$(-0.0560047, -0.0700059)$	$(-0.0543238, -0.0676943)$
Paired difference	4	S	$(-0.0560047, -0.0700059)$	$(-0.0559262, -0.0698957)$

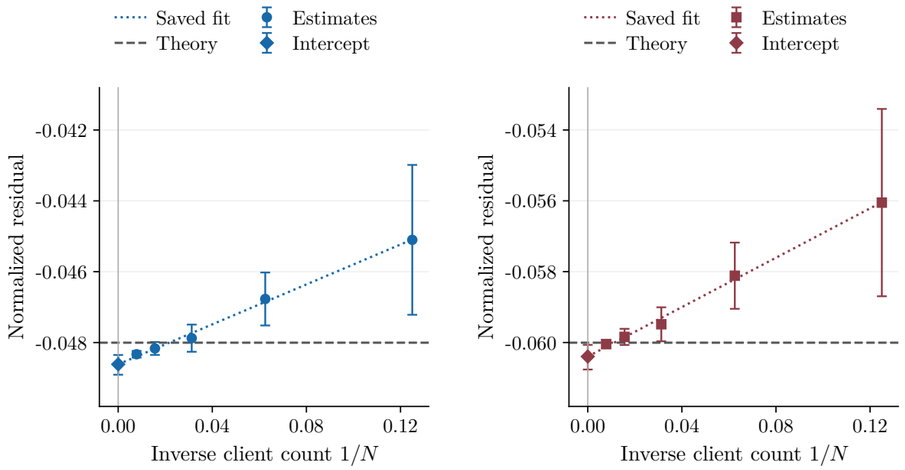
Results and interpretation. Figure 1 displays the matched estimates and analytical targets. The $H = 2$ intercept intervals contain both targets. For $H = 4$, the smaller-step grid reduces absolute intercept errors by about 95%, although its intervals still exclude the targets. The experiment therefore resolves a nonzero control-associated contribution and shows improved coefficient agreement on the finer grid, with a remaining $H = 4$ offset. Appendix C.2.2 examines this offset: the grid-point residuals scale approximately with $\gamma + 1/N$, and their finite-step, finite-client, and finite-run sources remain unresolved.



(a) Two local steps

(b) Four local steps

Figure 1: Matched SCAFFOLD–FedAvg coefficient estimates for (a) $H = 2$ and (b) $H = 4$ on the primary joint grids. The ordinate is $\widehat{D}_j$ from Equation (34). Circles and squares show the two coordinates, dotted lines show finite-grid linear fits, and dashed horizontal lines give the analytical targets. Open diamonds at $\delta = 0$ mark the fitted intercepts. Pointwise bars use simultaneous Student-t intervals; intercept bars use simultaneous seed-block-bootstrap intervals.



(a) Coordinate 1

(b) Coordinate 2

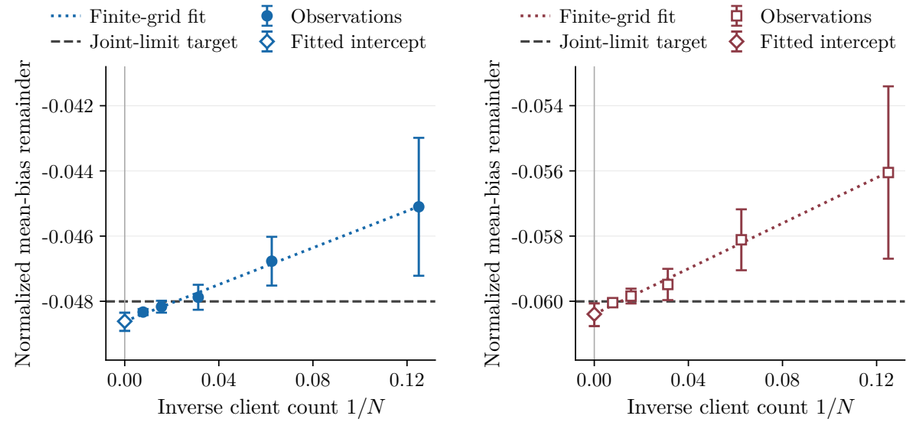
Figure 2: Client-count dependence of the normalized SCAFFOLD mean bias at fixed step size, shown by coordinate. Horizontal references give $B_{20}^{\mathrm{SCAFFOLD}}(2)$; the fixed-step expansion also permits an $O(\gamma)$ offset. Bars show simultaneous Student-t intervals at grid points and seed-block-bootstrap intervals for fitted intercepts.

state translation to floating-point accuracy. This illustrates how correction of affine gradient offsets can coexist with the same stochastic stationary-bias mechanism.

6.2.3 Client-count dependence at fixed step size

Design and rationale. The joint-grid experiment changes N and γ together. Here, we hold the step size and local-step count fixed and increase N, isolating the effect of additional client averaging. A client-independent second-order term should remain visible as finite-client corrections to the normalized statistic decrease.

Results and interpretation. The normalized SCAFFOLD contribution remains visibly nonzero across the tested client counts (Figure 2). The fitted offsets from the joint-limit coefficient are reported in Table 13; the fixed-step expansion permits an $O(\gamma)$ offset. This result illustrates the distinction between increasing the client count and reducing the step size: additional clients suppress the client-dependent term while leaving a residual mean shift.



(a) Coordinate 1

(b) Coordinate 2

Figure 2: Client-count dependence at fixed step size, shown by coordinate. The ordinate is $\widehat{\mathcal{C}}_j^{\mathrm{SCAFFOLD}}$ from Equation (33): the mean-bias estimate after subtracting the shared γ/N term and dividing by γ^2. Dotted lines give finite-grid fits and open diamonds their intercepts at $1/N = 0$. Dashed horizontal lines give the joint-limit references $B_{20}^{\mathrm{SCAFFOLD}}(2)$; a fixed step size permits an $O(\gamma)$ offset. Bars show simultaneous Student-t intervals at grid points and seed-block-bootstrap intervals for fitted intercepts.

6.2.4 Multidimensional and coordinate consistency

Design and rationale. A five-dimensional SCAFFOLD instance tests the coefficient formula beyond the primary two-dimensional example, again using noncommuting Hessian and covariance matrices. A separate rotation check transforms the objective, noise, and initial state together. The resulting trajectories and coefficients should transform by the same orthogonal map.

Results and interpretation. All ten coordinate targets across $H = 2$ and $H = 4$ lie within the reported simultaneous intervals (Table 14). The rotation check has pathwise residuals below 1.6×10^{-14} (Appendix C.3.2). The five-dimensional results support the coefficient formula across coordinates; the rotation check confirms that the implementation respects its coordinate transformation.

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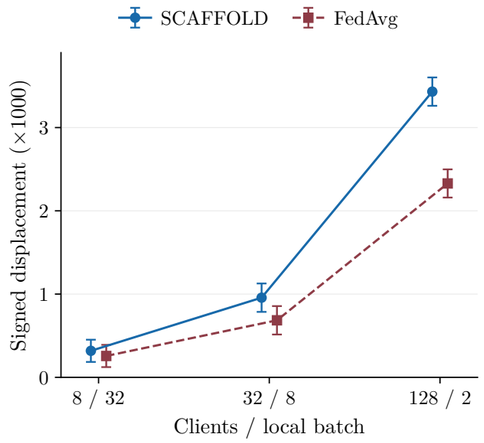


Figure 3: Finite-tail mean displacement under a fixed sampling budget. Each category gives a client-count/minibatch allocation (N, m) with $Nm = 256$. The ordinate is the signed projection in Equation (35), scaled by 10^3. Bars are nominal 95% run-seed intervals; small offsets distinguish the algorithms at the same allocation.

covariance is $\Sigma_{\text{ref}}/(Nm)$, where Σ_{ref} is the single-example gradient-noise covariance. The local covariance is instead Σ_{ref}/m, whose substitution into Equation (21) suggests a γ^2/m dependence. This motivates comparing allocations that have equal reference aggregate noise and different local noise levels.

We measure finite-tail displacement along the preselected unit direction $v \propto -A_{\text{ref}}^{-1} T_{\text{ref}}[\Sigma_{\text{ref}}]$, with $A_{\text{ref}} = \nabla^2 f(x_{\text{ref}})$ and $T_{\text{ref}} = \nabla^3 f(x_{\text{ref}})$:

$$\widehat{\delta}_v^{\mathcal{A}} = v^\top (\overline{x}^{\mathcal{A}} - x_{\text{ref}}), \tag{35}$$

Here $\overline{x}^{\mathcal{A}}$ averages the per-run tail means across independent runs. The statistic measures the signed mean displacement along a direction fixed before the training runs.

Results and interpretation. Displacement increases as the same sample budget is distributed among more clients with smaller local batches (Figure 3). The $N = 128$ minus $N = 8$ contrasts are 0.003114 for SCAFFOLD and 0.002072 for FedAvg, with adjusted intervals $[0.002749, 0.003478]$ and $[0.001713, 0.002431]$. Raw variance estimates remain near 0.00235–0.00236. Equal reference aggregate covariance therefore coexists with different finite-tail mean displacements: the local allocation carries information that server-level covariance alone does not capture.

The training-configuration comparison in Table 5 provides a complementary interpretation. Reducing γ lowers both displacement and raw variance; reducing H lowers displacement with much less variance change. These are comparisons under the run schedules in Table 8. The disjoint IID and label-skew results in Appendix C.4.1 additionally show how heterogeneity-induced drift correction can dominate the algorithm difference when the common-objective condition is relaxed.

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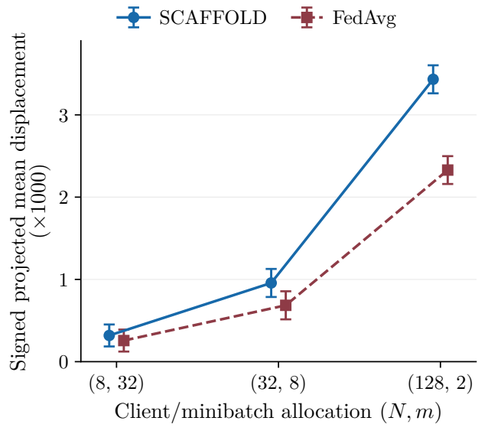


Figure 3: Finite-tail mean displacement under a fixed sampling budget. Each category gives a client-count/minibatch allocation (N, m) with $Nm = 256$. The ordinate is the signed projection in Equation (35), scaled by 10^3. Bars are nominal 95% run-seed intervals; small offsets distinguish the algorithms at the same allocation.

Table 5: Step-size and local-step comparisons for the shared objective at $N = 32, m = 8$. All entries are multiplied by 10^3. $\widehat{\delta}_v$ is signed finite-tail mean displacement and $\widehat{V}$ is raw-checkpoint parameter variance. S: SCAFFOLD; F: FedAvg.

Setting	$\widehat{\delta}_v^{\text{S}}$	$\widehat{\delta}_v^{\text{F}}$	$\widehat{V}^{\text{S}}$	$\widehat{V}^{\text{F}}$
$H = 4,\ \gamma = \gamma_b$	0.9567	0.6851	2.3621	2.3613
$H = 4,\ \gamma = \gamma_b/2$	0.3547	0.2796	1.1723	1.1722
$H = 4,\ \gamma = \gamma_b/4$	0.0622	0.0420	0.5860	0.5860
$H = 2,\ \gamma = \gamma_b$	0.3402	0.2474	2.3601	2.3599

6.4 Model averaging changes the error decomposition

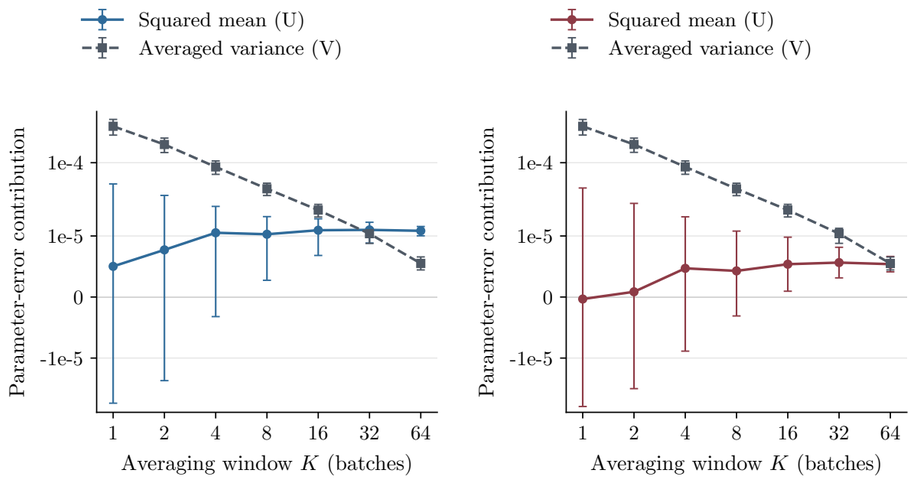
Design and rationale. Under stationarity, temporal averaging preserves the expected model while potentially reducing variance. We therefore examine whether mean displacement becomes a larger share of parameter error when successive global models are averaged. We use the same shared-objective runs and average successive global models within each run over nested windows. Let $Y_{r,K}^{\mathcal{A}}$ be run r's output averaged over its last K equal-length batches. Independent runs supply repeated outputs, while the windows share a common endpoint. Table 20 gives the window lengths and computational budget.

For a random output $Y_K^{\mathcal{A}}$ of this finite-window procedure,

$$\mathbb{E}\|Y_K^{\mathcal{A}} - x_{\text{ref}}\|^2 = \|\mathbb{E}[Y_K^{\mathcal{A}}] - x_{\text{ref}}\|^2 + \operatorname{tr} \operatorname{Cov}(Y_K^{\mathcal{A}}). \tag{36}$$

We estimate squared mean displacement by the sampling-corrected cross-run statistic $\widehat{U}_K$ and averaged-output variance by $\widehat{V}_K$. Their sum is the estimated parameter MSE. Sampling correction allows $\widehat{U}_K$ to be negative with finitely many runs; the plots retain these values. Appendix C.4.3 gives the estimators.

Results and interpretation. In the primary allocation, mean displacement accounts for only about 0.50% and 0.23% of raw-checkpoint parameter MSE for SCAFFOLD and FedAvg,



(a) SCAFFOLD (b) FedAvg

Figure 4: Parameter-error decomposition after temporal model averaging. The panels compare estimated squared mean displacement $\widehat{U}_K$ with averaged-output variance $\widehat{V}_K$ over windows of K batches in the primary shared-objective configuration. Each batch contains 1,280 rounds. Bars are approximate simultaneous 95% primary-family intervals. The symmetric-log axis retains negative sampling-corrected estimates and intervals crossing zero.

6.5 Directional agreement beyond the additive-noise model

Design and rationale. We examine whether the curvature–noise geometry helps explain the algorithmic mean displacement under natural minibatch noise. The post-hoc comparison substitutes the reference-objective quantities from Section 6.3 into the analytical expression:

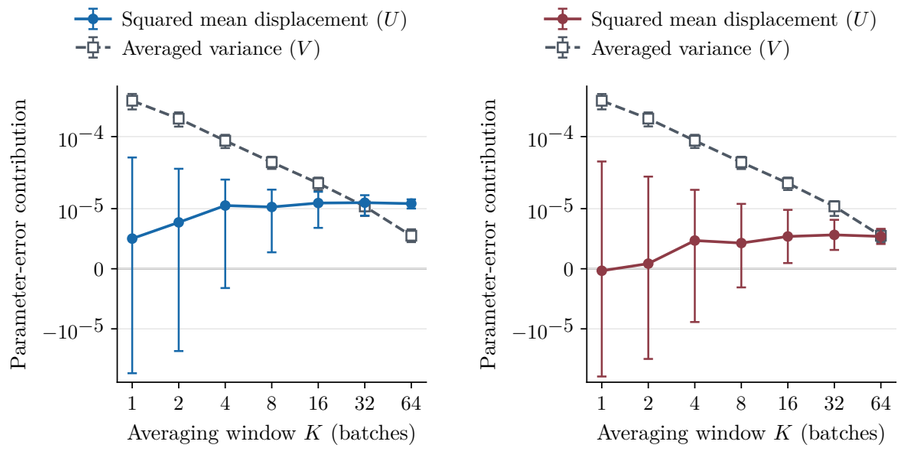
$$p_{\gamma,H,m} = -\frac{\gamma^2}{m}\,\kappa(H)\,A_{\mathrm{ref}}^{-1}T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}], \qquad \kappa(H) = \frac{(H-1)(2H-1)}{12H}. \tag{37}$$

The observed difference is $\widehat{\Delta}_K = R^{-1}\sum_r (Y_{r,K}^{\mathrm{SCAFFOLD}} - Y_{r,K}^{\mathrm{FedAvg}})$ over $R = 12$ paired runs. The proxy is computed directly from the objective and reference covariance. Comparing all 124 coordinates tests its geometric alignment with the observed mean difference. Cosine measures directional agreement; projection ratio and relative residual measure amplitude agreement and vector mismatch. Because both algorithms' second-order terms share the same analytical direction, the amplitude and local-step comparisons provide information beyond directional alignment.

Results and interpretation. At the designated longest window, the six nonzero-proxy configurations have cosines 0.9965–0.9993, projection ratios 0.716–0.938, and relative residuals 0.089–0.291 (Table 6). All six proxy norms exceed the observed norms. At fixed H, N, m, reducing γ moves the projection ratio toward one and decreases the residual. Alternative fixed directions have lower cosine estimates (Appendix C.4.4). The analytical geometry thus describes the observed direction more accurately than its amplitude.

The local-step comparison further separates these two aspects of agreement: at the base step size and $N = 32, m = 8$, the observed $H = 4/H = 2$ projected ratio is 2.927, compared with

respectively. Figure 4 shows how this composition changes after averaging. From $K = 1$ to 64, averaged-output variance decreases about 57-fold. At $K = 64$, the mean-displacement shares reach 68.1% and 49.5%, with simultaneous intervals $[62.6\%, 73.7\%]$ and $[41.1\%, 57.9\%]$. Systematic displacement therefore becomes a substantial part of the averaged-model error in this experiment, even though individual checkpoints are variance-dominated.



(a) SCAFFOLD (b) FedAvg

Figure 4: Parameter-error decomposition after temporal model averaging. The panels compare estimated squared mean displacement $\widehat{U}_K$ with averaged-output variance $\widehat{V}_K$ over windows of K batches in the primary shared-objective configuration. Each batch contains 1,280 rounds. Bars are approximate simultaneous 95% primary-family intervals. The symmetric-log axis is linear between -10^{-5} and 10^{-5} and retains negative sampling-corrected estimates and intervals crossing zero.

At the longest window, the SCAFFOLD-minus-FedAvg parameter-MSE contrast is 6.39×10^{-6}, with interval $[5.83, 6.95] \times 10^{-6}$, mainly associated with squared mean displacement. The corresponding training-gap contrast is 3.64×10^{-7}, while SCAFFOLD has lower held-out log loss by 3.27×10^{-5}. Thus parameter error and predictive loss respond differently. The averaging result concerns finite-tail outputs; Appendix C.4.3 reports the remaining temporal uncertainty.

6.5 Directional agreement beyond the additive-noise model

Design and rationale. We examine whether the curvature–noise geometry helps explain the algorithmic mean displacement under natural minibatch noise. The post-hoc comparison substitutes the reference-objective quantities from Section 6.3 into the analytical expression:

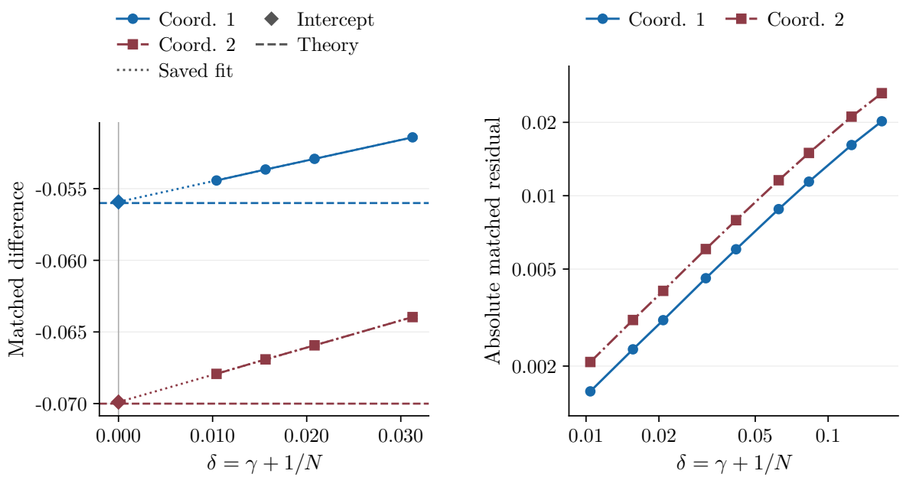
$$p_{\gamma,H,m} = -\frac{\gamma^2}{m}\,\kappa(H)\,A_{\mathrm{ref}}^{-1}T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}], \qquad \kappa(H) = \frac{(H-1)(2H-1)}{12H}. \tag{37}$$

The observed difference is $\widehat{\Delta}_K = R^{-1}\sum_r (Y_{r,K}^{\mathrm{SCAFFOLD}} - Y_{r,K}^{\mathrm{FedAvg}})$ over $R = 12$ paired runs. The proxy is computed directly from the objective and reference covariance. Comparing all 124 coordinates tests its geometric alignment with the observed mean difference. Cosine measures directional agreement; projection ratio and relative residual measure amplitude agreement and vector mismatch. Because both algorithms' second-order terms share the same analytical

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Table 12: Matched $H = 4$ grid-point residuals against $\delta = \gamma + 1/N$. The last two columns summarize the observed residual scale across the combined grid.

j	$\lvert e_\Delta \rvert$ at $\delta_{\min}$	$\lvert e_\Delta \rvert$ at $\delta_{\max}$	$\lvert e_\Delta \rvert / \delta$ range	Log–log slope
1	0.0015750	0.0202050	$[0.121, 0.151]$	0.927
2	0.0020785	0.0263460	$[0.158, 0.200]$	0.923



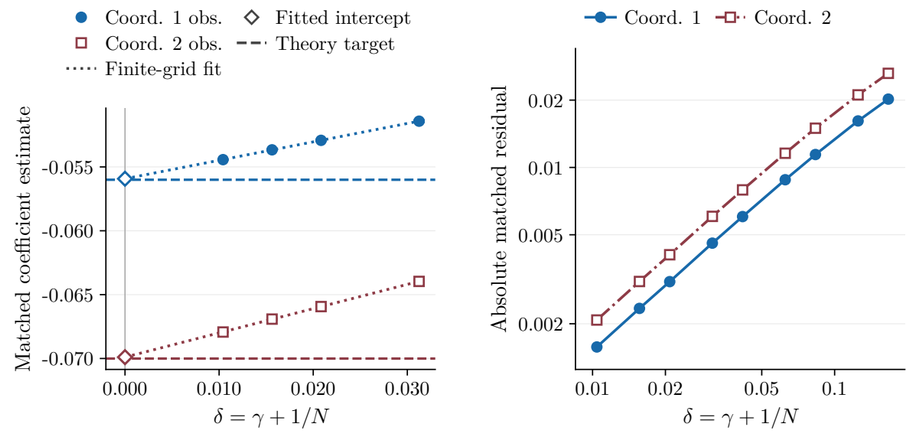
(a) Smaller-step path (b) Absolute residuals

Figure 5: Matched $H = 4$ finite-grid sensitivity. (a) Smaller-step estimates and their linear extrapolations. (b) Absolute residuals versus $\delta = \gamma + 1/N$ across all nine points. The scaled residuals and log–log slopes are given in Table 12. Pointwise bars in (a) use simultaneous Student-t intervals; intercept bars use seed-block-bootstrap intervals.

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Table 12: Matched $H = 4$ grid-point residuals against $\delta = \gamma + 1/N$. The last two columns summarize the observed residual scale across the combined grid.

j	$\lvert e_\Delta \rvert$ at $\delta_{\min}$	$\lvert e_\Delta \rvert$ at $\delta_{\max}$	$\lvert e_\Delta \rvert / \delta$ range	Log–log slope
1	0.0015750	0.0202050	$[0.121, 0.151]$	0.927
2	0.0020785	0.0263460	$[0.158, 0.200]$	0.923



(a) Smaller-step path (b) Absolute residuals

Figure 5: Matched $H = 4$ finite-grid sensitivity. (a) Smaller-step estimates (circles and squares), finite-grid linear fits (dotted), and analytical targets (dashed). Open diamonds at $\delta = 0$ mark the fitted intercepts. (b) Absolute residuals versus $\delta = \gamma + 1/N$ across all nine points on logarithmic axes; lines connect the observed residuals. The scaled residuals and log–log slopes are given in Table 12. Pointwise bars in (a) use simultaneous Student-t intervals; intercept bars use seed-block-bootstrap intervals.

Table 13: Sampling intervals and offsets for the fixed-step extrapolated intercepts in Figure 2.

j	95% simultaneous CI	Intercept − theory
1	$[-0.0489079, -0.0483504]$	-0.0006102
2	$[-0.0607622, -0.0600656]$	-0.0003902

C.2.3 Fixed-step and five-dimensional results

Design and rationale. These studies separately examine client count and dimension using the objective family in Equation (74). The fixed-step study uses the grid in Table 7. The five-dimensional instance has positive-definite A, C with $\|AC - CA\|_F = 0.0876171216$.

Results and interpretation. Table 13 adds the fixed-step intercept intervals and offsets to Figure 2. Table 14 supplies all five coordinates for each local-step count. These details support the client-count and multidimensional interpretations in Sections 6.2.3 and 6.2.4.

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Table 2: Analytical coefficient vectors and fitted finite-grid intercepts. Individual rows target $B_{20}^{\mathcal{A}}(H)$; matched-pair rows target $B_{20}^{\mathrm{ctrl}}(H)$. P and S denote the primary and smaller-step joint grids in Table 7. Values are rounded to seven decimal places; simultaneous intervals are in Table 11.

Study	H	Grid	Target vector	Estimated intercept vector
SCAFFOLD	2	P	$(-0.0480040, -0.0600050)$	$(-0.0476713, -0.0595811)$
SCAFFOLD	4	P	$(-0.1520127, -0.1900159)$	$(-0.1502614, -0.1874524)$
SCAFFOLD	4	S	$(-0.1520127, -0.1900159)$	$(-0.1514772, -0.1893273)$
FedAvg	2	P	$(-0.0320027, -0.0400033)$	$(-0.0320430, -0.0400539)$
FedAvg	4	P	$(-0.0960080, -0.1200100)$	$(-0.0959717, -0.1198059)$
FedAvg	4	S	$(-0.0960080, -0.1200100)$	$(-0.0959727, -0.1199590)$
Paired difference	2	P	$(-0.0160013, -0.0200017)$	$(-0.0160363, -0.0200366)$
Paired difference	4	P	$(-0.0560047, -0.0700059)$	$(-0.0543238, -0.0676943)$
Paired difference	4	S	$(-0.0560047, -0.0700059)$	$(-0.0559262, -0.0698957)$

Results and interpretation. Figure 1 displays the matched estimates and analytical targets. The $H = 2$ intercept intervals contain both targets. For $H = 4$, the smaller-step grid reduces absolute intercept errors by about 95%, although its intervals still exclude the targets. The experiment therefore resolves a nonzero control-associated contribution and shows improved coefficient agreement on the finer grid, with a remaining $H = 4$ offset. Appendix C.2.2 examines this offset: the grid-point residuals scale approximately with $\gamma + 1/N$, and their finite-step, finite-client, and finite-run sources remain unresolved.

6.2.3 Client-count dependence at fixed step size

Design and rationale. The joint-grid experiment changes N and γ together. Here, we hold the step size and local-step count fixed and increase N, isolating the effect of additional client averaging. A client-independent second-order term should remain visible as finite-client corrections to the normalized statistic decrease.

Results and interpretation. The normalized SCAFFOLD contribution remains visibly nonzero across the tested client counts (Figure 2). The fitted offsets from the joint-limit coefficient are reported in Table 13; the fixed-step expansion permits an $O(\gamma)$ offset. This result illustrates the distinction between increasing the client count and reducing the step size: additional clients suppress the client-dependent term while leaving a residual mean shift.

6.2.4 Multidimensional and coordinate consistency

Design and rationale. A five-dimensional SCAFFOLD instance tests the coefficient formula beyond the primary two-dimensional example, again using noncommuting Hessian and covariance matrices. A separate rotation check transforms the objective, noise, and initial state together. The resulting trajectories and coefficients should transform by the same orthogonal map.

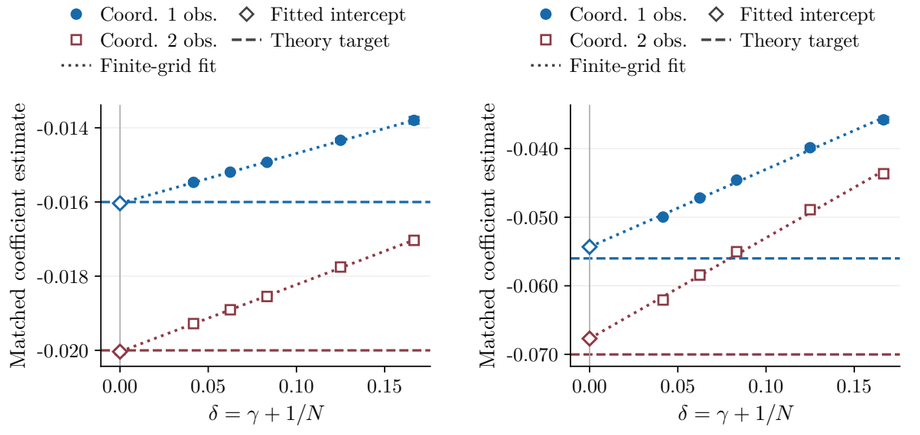
Results and interpretation. All ten coordinate targets across $H = 2$ and $H = 4$ lie within the reported simultaneous intervals (Table 14). The rotation check has pathwise residuals below 1.6×10^{-14} (Appendix C.3.2). The five-dimensional results support the coefficient formula across coordinates; the rotation check confirms that the implementation respects its coordinate transformation.

AFTER | final page 11

Table 2: Analytical coefficient vectors and fitted finite-grid intercepts. Individual rows target $B_{20}^{\mathcal{A}}(H)$; matched-pair rows target $B_{20}^{\mathrm{ctrl}}(H)$. P and S denote the primary and smaller-step joint grids in Table 7. Values are rounded to seven decimal places; simultaneous intervals are in Table 11.

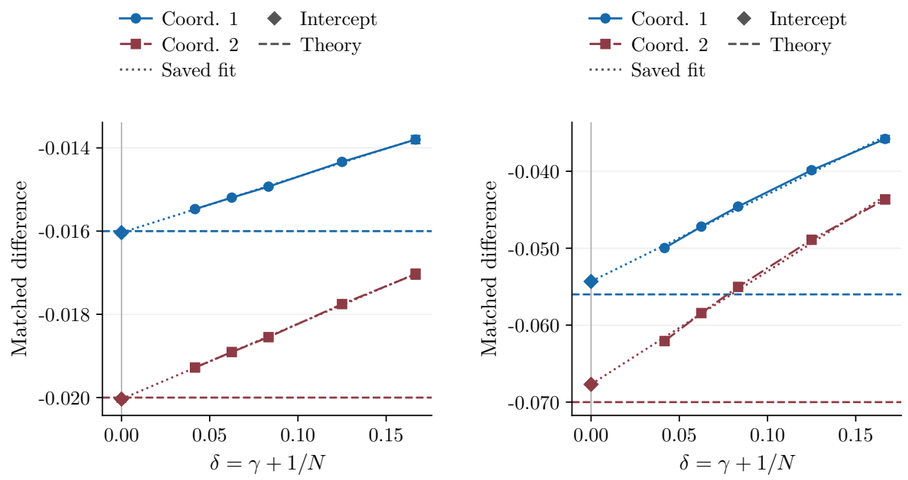
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Results and interpretation. Figure 1 displays the matched estimates and analytical targets. The $H = 2$ intercept intervals contain both targets. For $H = 4$, the smaller-step grid reduces absolute intercept errors by about 95%, although its intervals still exclude the targets. The experiment therefore resolves a nonzero control-associated contribution and shows improved coefficient agreement on the finer grid, with a remaining $H = 4$ offset. Appendix C.2.2 examines this offset: the grid-point residuals scale approximately with $\gamma + 1/N$, and their finite-step, finite-client, and finite-run sources remain unresolved.



(a) Two local steps (b) Four local steps

Figure 1: Matched SCAFFOLD–FedAvg coefficient estimates for (a) $H = 2$ and (b) $H = 4$ on the primary joint grids. The ordinate is $\widehat{D}_j$ from Equation (34). Circles and squares show the two coordinates, dotted lines show finite-grid linear fits, and dashed horizontal lines give the analytical targets. Open diamonds at $\delta = 0$ mark the fitted intercepts. Pointwise bars use simultaneous Student-t intervals; intercept bars use simultaneous seed-block-bootstrap intervals.



(a) Two local steps

(b) Four local steps

Figure 1: Matched SCAFFOLD–FedAvg coefficient estimates for (a) $H = 2$ and (b) $H = 4$ on the primary joint grids. The ordinate is $\widehat{D}_j$ from Equation (34); dashed lines give the analytical targets and dotted curves give linear extrapolations. Pointwise bars use simultaneous Student-t intervals; intercept bars use simultaneous seed-block-bootstrap intervals.

6.2.5 Local-step dependence on an independent grid

Design and rationale. The theory assigns different roles to local training length: B_{11} is shared across H, whereas $B_{20}^{\mathrm{ctrl}}(H)$ changes explicitly with H. An independent implementation tests both predictions. Individual fits of $(N/\gamma)\widehat{b}$ estimate B_{11} freely. Matched fits on a Cartesian grid vary γ and $1/N$ independently, allowing a direct comparison with the predicted relation $B_{20}^{\mathrm{ctrl}}(4) = 3.5B_{20}^{\mathrm{ctrl}}(2)$.

Results and interpretation. The leading-coefficient intervals contain their targets for $H = 1, 2, 4$; both algorithms have offsets at $H = 8$. The adjusted SCAFFOLD $H = 8$ versus $H = 2$ leading-intercept contrasts exclude zero (Table 15). For the matched coefficient, Table 3 shows target inclusion at $H = 2, 4$ and target exclusion at $H = 8$. The adjusted $H = 4$ minus $3.5(H = 2)$ contrast contains zero in both coordinates. These independent estimates support the predicted local-step relation at $H = 2, 4$ and identify $H = 8$ as more sensitive to the fitted grid. Restricted-grid estimates and the separate joint-grid $H = 4$ discrepancy are documented in Appendix C.2.

6.2.6 Affine-shift equivalence

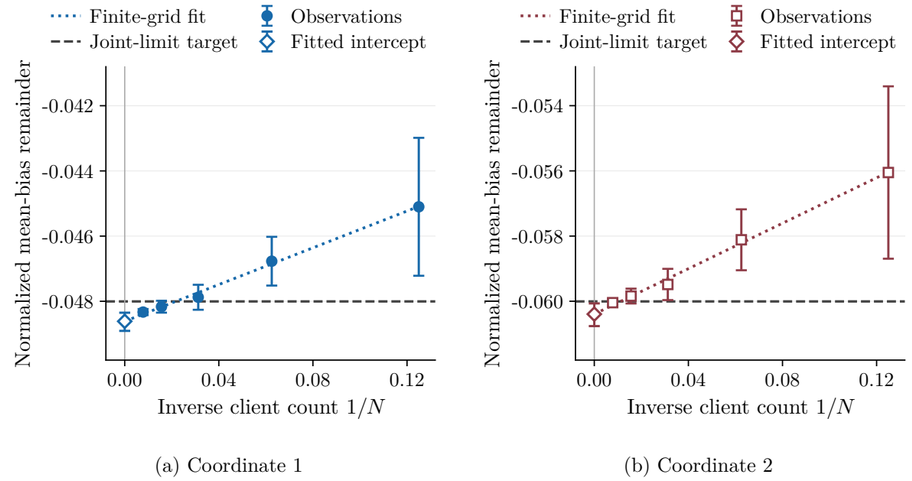
Design and rationale. Proposition 4.4 predicts exact equivalence after client-specific gradient offsets are absorbed into the controls. We test this identity by pairing affine-shift and homogeneous recursions with translated initial controls and identical noise, across several shift sizes, client counts, and local-step counts.

Results and interpretation. Across 24 configurations of 2,000 rounds, the maximum control-state residual is 1.64313×10^{-14}. The numerical recursions therefore reproduce the predicted

6.2.3 Client-count dependence at fixed step size

Design and rationale. The joint-grid experiment changes N and γ together. Here, we hold the step size and local-step count fixed and increase N, isolating the effect of additional client averaging. A client-independent second-order term should remain visible as finite-client corrections to the normalized statistic decrease.

Results and interpretation. The normalized SCAFFOLD contribution remains visibly nonzero across the tested client counts (Figure 2). The fitted offsets from the joint-limit coefficient are reported in Table 13; the fixed-step expansion permits an $O(\gamma)$ offset. This result illustrates the distinction between increasing the client count and reducing the step size: additional clients suppress the client-dependent term while leaving a residual mean shift.



(a) Coordinate 1

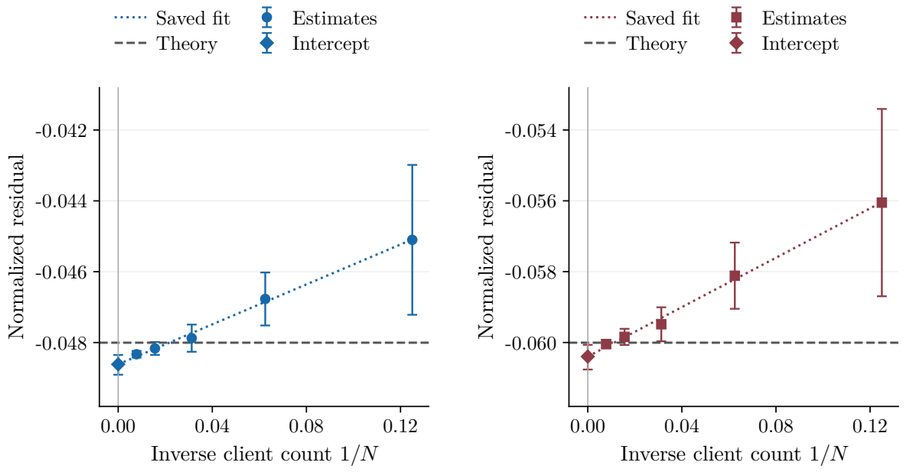
(b) Coordinate 2

Figure 2: Client-count dependence at fixed step size, shown by coordinate. The ordinate is $\widehat{\mathcal{C}}_j^{\mathrm{SCAFFOLD}}$ from Equation (33): the mean-bias estimate after subtracting the shared γ/N term and dividing by γ^2. Dotted lines give finite-grid fits and open diamonds their intercepts at $1/N = 0$. Dashed horizontal lines give the joint-limit references $B_{20}^{\mathrm{SCAFFOLD}}(2)$; a fixed step size permits an $O(\gamma)$ offset. Bars show simultaneous Student-t intervals at grid points and seed-block-bootstrap intervals for fitted intercepts.

6.2.4 Multidimensional and coordinate consistency

Design and rationale. A five-dimensional SCAFFOLD instance tests the coefficient formula beyond the primary two-dimensional example, again using noncommuting Hessian and covariance matrices. A separate rotation check transforms the objective, noise, and initial state together. The resulting trajectories and coefficients should transform by the same orthogonal map.

Results and interpretation. All ten coordinate targets across $H = 2$ and $H = 4$ lie within the reported simultaneous intervals (Table 14). The rotation check has pathwise residuals below 1.6×10^{-14} (Appendix C.3.2). The five-dimensional results support the coefficient formula across coordinates; the rotation check confirms that the implementation respects its coordinate transformation.



(a) Coordinate 1

(b) Coordinate 2

Figure 2: Client-count dependence of the normalized SCAFFOLD mean bias at fixed step size, shown by coordinate. Horizontal references give $B_{20}^{\mathrm{SCAFFOLD}}(2)$; the fixed-step expansion also permits an $O(\gamma)$ offset. Bars show simultaneous Student-t intervals at grid points and seed-block-bootstrap intervals for fitted intercepts.

state translation to floating-point accuracy. This illustrates how correction of affine gradient offsets can coexist with the same stochastic stationary-bias mechanism.

Table 3: Control-coefficient estimates on the independent Cartesian grid. Each row compares the fitted intercept with $B_{20}^{\mathrm{ctrl}}(H)$. Intervals are nominal 95% run-seed intervals.

H	j	Estimate	Target	Nominal interval
2	1	−0.01599546	−0.01600134	[−0.01601159, −0.01597933]
2	2	−0.01999408	−0.02000167	[−0.02001422, −0.01997394]
4	1	−0.05595226	−0.05600469	[−0.05601109, −0.05589342]
4	2	−0.06993656	−0.07000586	[−0.07000999, −0.06986314]
8	1	−0.13962104	−0.14001171	[−0.13983973, −0.13940235]
8	2	−0.17449119	−0.17501464	[−0.17476345, −0.17421894]

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Results and interpretation. Across 24 configurations of 2,000 rounds, the maximum control-state residual is 1.64313×10^{-14}. The numerical recursions therefore reproduce the predicted state translation to floating-point accuracy. This illustrates how correction of affine gradient offsets can coexist with the same stochastic stationary-bias mechanism.

6.2.7 Contraction cancellation and covariance-matched noise

Design and rationale. These tests isolate the specific role of $T[C]$. A nonquadratic trigonometric objective has $T \neq 0$, and the covariance family $C_\rho = \mathrm{diag}(1 + \rho, 1 - \rho)$ preserves total noise variance. At $\rho = 0$, the contraction vanishes while $B_{11} = (-0.0125, 0)^\top$ remains nonzero, isolating cancellation of the client-independent coefficient. The positive and negative covariance controls produce opposite-signed targets. At one fixed covariance, bounded symmetric and asymmetric noise laws change third and fourth moments while preserving the predicted coefficients. Unconstrained Cartesian-grid fits estimate the individual and matched quantities; the model and laws are given in Appendix C.1.4.

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Results and interpretation. Table 4 separates the matched null from both nonzero covariance controls and recovers their predicted signs and magnitudes. The noise-law contrast is small. In both coordinates, the null and law-contrast intervals fall within the prespecified coefficient-resolution margin ±0.004375, equal to 20% of the nonzero matched target magnitude. Thus the matched increment responds to covariance geometry and agrees across covariance-matched noise laws at the reported resolution. Individual first-coordinate null and law contrasts remain imprecise; one small paired second-coordinate interval excludes zero while lying within the resolution margin. Appendix C.2.5 gives these complementary outcomes and their sensitivity analyses.

Table 4: Contraction and noise-law checks for the first coordinate of the matched coefficient at $H = 4$. The covariance controls predict positive, zero, and negative coefficients; the last row compares noise laws with identical covariance. Intervals are approximate simultaneous 95% sampling intervals for the prespecified 36-scalar family.

ρ	Noise law	Target	Estimate	Simultaneous interval
−1/4	Symmetric	0.021875	0.02187858	[0.02184128, 0.02191589]
0	Symmetric	0.000000	−0.00002056	[−0.00006150, 0.00002039]
+1/4	Symmetric	−0.021875	−0.02185115	[−0.02189536, −0.02180693]
+1/4	Asymmetric	−0.021875	−0.02188179	[−0.02193153, −0.02183205]
Symmetric minus asymmetric ($\rho = 1/4$)		0.000000	0.00003064	[−0.00003564, 0.00009692]

6.3 Real-data implications of a fixed sampling budget

Design and rationale. We examine whether equal aggregate noise makes client allocations interchangeable. All clients optimize the same a9a logistic-regression objective, and the total minibatch budget Nm is fixed. At the reference minimizer x_{ref}, the server-averaged noise

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Symmetric minus asymmetric ($\rho = 1/4$)		0.000000	0.00003064	[−0.00003564, 0.00009692]

6.3 Real-data implications of a fixed sampling budget

Design and rationale. We examine whether equal aggregate noise makes client allocations interchangeable. All clients optimize the same a9a logistic-regression objective, and the total minibatch budget Nm is fixed. At the reference minimizer x_{ref}, the server-averaged noise covariance is $\Sigma_{\mathrm{ref}}/(Nm)$, where Σ_{ref} is the single-example gradient-noise covariance. The local covariance is instead Σ_{ref}/m, whose substitution into Equation (21) suggests a γ^2/m dependence. This motivates comparing allocations that have equal reference aggregate noise and different local noise levels.

We measure finite-tail displacement along the preselected unit direction $v \propto -A_{\mathrm{ref}}^{-1} T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}]$, with $A_{\mathrm{ref}} = \nabla^2 f(x_{\mathrm{ref}})$ and $T_{\mathrm{ref}} = \nabla^3 f(x_{\mathrm{ref}})$:

$$\widehat{\delta}_v^{\mathcal{A}} = v^\top(\overline{x}^{\mathcal{A}} - x_{\mathrm{ref}}), \tag{35}$$

Here $\overline{x}^{\mathcal{A}}$ averages the per-run tail means across independent runs. The statistic measures the signed mean displacement along a direction fixed before the training runs.

Results and interpretation. Displacement increases as the same sample budget is distributed among more clients with smaller local batches (Figure 3). The $N = 128$ minus $N = 8$ contrasts are 0.003114 for SCAFFOLD and 0.002072 for FedAvg, with adjusted intervals $[0.002749, 0.003478]$ and $[0.001713, 0.002431]$. Raw variance estimates remain near 0.00235–0.00236. Equal reference aggregate covariance therefore coexists with different finite-tail mean displacements: the local allocation carries information that server-level covariance alone does not capture.

The training-configuration comparison in Table 5 provides a complementary interpretation. Reducing γ lowers both displacement and raw variance; reducing H lowers displacement with much less variance change. These are comparisons under the run schedules in Table 8. The disjoint IID and label-skew results in Appendix C.4.1 additionally show how heterogeneity-induced drift correction can dominate the algorithm difference when the common-objective condition is relaxed.

Table 5: Step-size and local-step comparisons for the shared objective at $N = 32, m = 8$. All entries are multiplied by 10^3. $\widehat{\delta}_v$ is signed finite-tail mean displacement and $\widehat{V}$ is raw-checkpoint parameter variance. S: SCAFFOLD; F: FedAvg.

Setting	$\widehat{\delta}_v^{\mathrm{S}}$	$\widehat{\delta}_v^{\mathrm{F}}$	$\widehat{V}^{\mathrm{S}}$	$\widehat{V}^{\mathrm{F}}$
$H = 4,\ \gamma = \gamma_b$	0.9567	0.6851	2.3621	2.3613
$H = 4,\ \gamma = \gamma_b/2$	0.3547	0.2796	1.1723	1.1722
$H = 4,\ \gamma = \gamma_b/4$	0.0622	0.0420	0.5860	0.5860
$H = 2,\ \gamma = \gamma_b$	0.3402	0.2474	2.3601	2.3599

6.4 Model averaging changes the error decomposition

Design and rationale. Under stationarity, temporal averaging preserves the expected model while potentially reducing variance. We therefore examine whether mean displacement becomes a larger share of parameter error when successive global models are averaged. We use the same shared-objective runs and average successive global models within each run over nested windows. Let $Y_{r,K}^{\mathcal{A}}$ be run r's output averaged over its last K equal-length batches. Independent runs supply repeated outputs, while the windows share a common endpoint. Table 20 gives the window lengths and computational budget.

For a random output $Y_K^{\mathcal{A}}$ of this finite-window procedure,

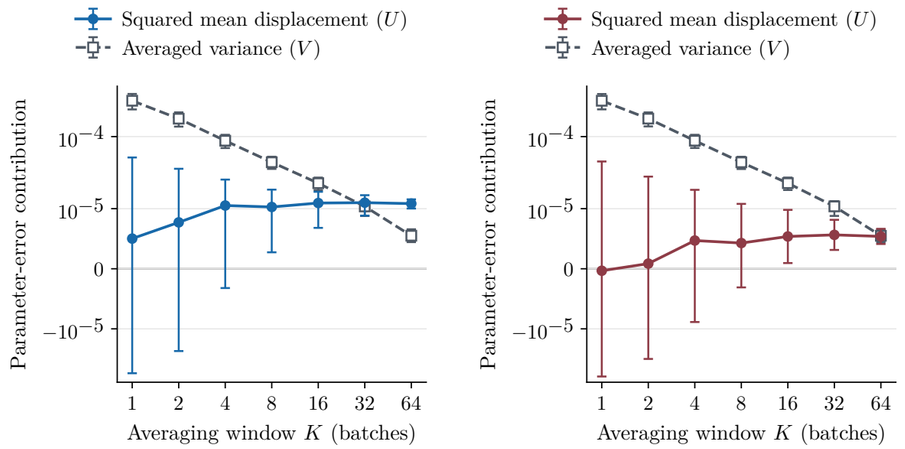
$$\mathbb{E}\|Y_K^{\mathcal{A}} - x_{\mathrm{ref}}\|^2 = \|\mathbb{E}[Y_K^{\mathcal{A}}] - x_{\mathrm{ref}}\|^2 + \operatorname{tr}\operatorname{Cov}(Y_K^{\mathcal{A}}). \tag{36}$$

We estimate squared mean displacement by the sampling-corrected cross-run statistic $\widehat{U}_K$ and averaged-output variance by V_K. Their sum is the estimated parameter MSE. Sampling correction allows $\widehat{U}_K$ to be negative with finitely many runs; the plots retain these values. Appendix C.4.3 gives the estimators.

Results and interpretation. In the primary allocation, mean displacement accounts for only about 0.50% and 0.23% of raw-checkpoint parameter MSE for SCAFFOLD and FedAvg, respectively. Figure 4 shows how this composition changes after averaging. From $K = 1$ to 64, averaged-output variance decreases about 57-fold. At $K = 64$, the mean-displacement shares reach 68.1% and 49.5%, with simultaneous intervals $[62.6\%, 73.7\%]$ and $[41.1\%, 57.9\%]$. Systematic displacement therefore becomes a substantial part of the averaged-model error in this experiment, even though individual checkpoints are variance-dominated.

At the longest window, the SCAFFOLD-minus-FedAvg parameter-MSE contrast is 6.39×10^{-6}, with interval $[5.83, 6.95] \times 10^{-6}$, mainly associated with squared mean displacement. The corresponding training-gap contrast is 3.64×10^{-7}, while SCAFFOLD has lower held-out log loss by 3.27×10^{-5}. Thus parameter error and predictive loss respond differently. The averaging result concerns finite-tail outputs; Appendix C.4.3 reports the remaining temporal uncertainty.

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(a) SCAFFOLD (b) FedAvg

Figure 4: Parameter-error decomposition after temporal model averaging. The panels compare estimated squared mean displacement $\widehat{U}_K$ with averaged-output variance $\widehat{V}_K$ over windows of K batches in the primary shared-objective configuration. Each batch contains 1,280 rounds. Bars are approximate simultaneous 95% primary-family intervals. The symmetric-log axis is linear between -10^{-5} and 10^{-5} and retains negative sampling-corrected estimates and intervals crossing zero.

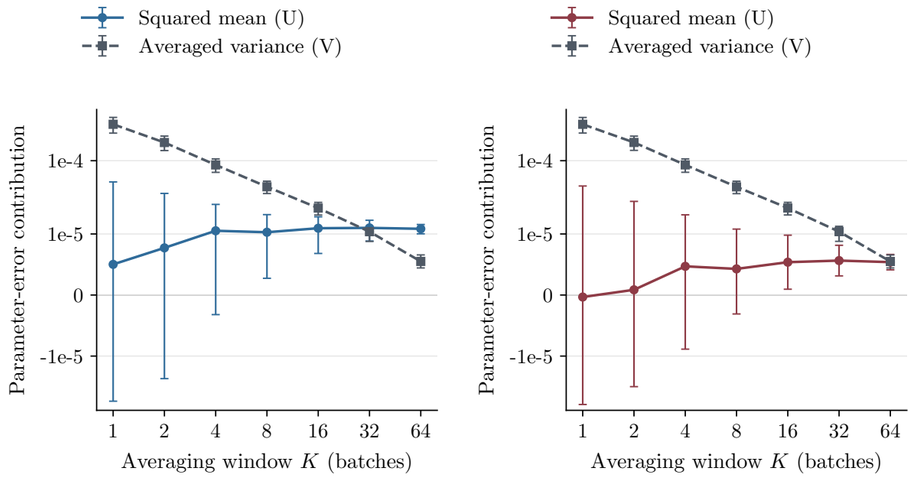
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6.5 Directional agreement beyond the additive-noise model

Design and rationale. We examine whether the curvature–noise geometry helps explain the algorithmic mean displacement under natural minibatch noise. The post-hoc comparison substitutes the reference-objective quantities from Section 6.3 into the analytical expression:

$$p_{\gamma,H,m} = -\frac{\gamma^2}{m}\,\kappa(H)\,A_{\mathrm{ref}}^{-1}T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}], \qquad \kappa(H) = \frac{(H-1)(2H-1)}{12H}. \tag{37}$$

The observed difference is $\widehat{\Delta}_K = R^{-1}\sum_r (Y_{r,K}^{\mathrm{SCAFFOLD}} - Y_{r,K}^{\mathrm{FedAvg}})$ over $R = 12$ paired runs. The proxy is computed directly from the objective and reference covariance. Comparing all 124 coordinates tests its geometric alignment with the observed mean difference. Cosine measures directional agreement; projection ratio and relative residual measure amplitude agreement and vector mismatch. Because both algorithms' second-order terms share the same analytical



(a) SCAFFOLD (b) FedAvg

Figure 4: Parameter-error decomposition after temporal model averaging. The panels compare estimated squared mean displacement $\widehat{U}_K$ with averaged-output variance $\widehat{V}_K$ over windows of K batches in the primary shared-objective configuration. Each batch contains 1,280 rounds. Bars are approximate simultaneous 95% primary-family intervals. The symmetric-log axis retains negative sampling-corrected estimates and intervals crossing zero.

6.5 Directional agreement beyond the additive-noise model

Design and rationale. We examine whether the curvature–noise geometry helps explain the algorithmic mean displacement under natural minibatch noise. The post-hoc comparison substitutes the reference-objective quantities from Section 6.3 into the analytical expression:

$$p_{\gamma,H,m} = -\frac{\gamma^2}{m}\,\kappa(H)\,A_{\mathrm{ref}}^{-1}T_{\mathrm{ref}}[\Sigma_{\mathrm{ref}}], \qquad \kappa(H) = \frac{(H-1)(2H-1)}{12H}. \tag{37}$$

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Results and interpretation. At the designated longest window, the six nonzero-proxy configurations have cosines 0.9965–0.9993, projection ratios 0.716–0.938, and relative residuals 0.089–0.291 (Table 6). All six proxy norms exceed the observed norms. At fixed H, N, m, reducing γ moves the projection ratio toward one and decreases the residual. Alternative fixed directions have lower cosine estimates (Appendix C.4.4). The analytical geometry thus describes the observed direction more accurately than its amplitude.

The local-step comparison further separates these two aspects of agreement: at the base step size and $N = 32, m = 8$, the observed $H = 4/H = 2$ projected ratio is 2.927, compared with

direction, the amplitude and local-step comparisons provide information beyond directional alignment.

Results and interpretation. At the designated longest window, the six nonzero-proxy configurations have cosines 0.9965–0.9993, projection ratios 0.716–0.938, and relative residuals 0.089–0.291 (Table 6). All six proxy norms exceed the observed norms. At fixed H, N, m, reducing γ moves the projection ratio toward one and decreases the residual. Alternative fixed directions have lower cosine estimates (Appendix C.4.4). The analytical geometry thus describes the observed direction more accurately than its amplitude.

Table 6: Post-hoc comparison of the analytical proxy $p = p_{\gamma,H,m}$ and the 124-dimensional finite-tail mean difference $\widehat{\Delta}_K$ at $K = 64$. Cosine $= p^\top\widehat{\Delta}_K/(\|p\|\|\widehat{\Delta}_K\|)$; projection ratio $= p^\top\widehat{\Delta}_K/\|p\|^2$; relative residual $= \|\widehat{\Delta}_K - p\|/\|p\|$. All six nonzero-proxy configurations are shown; the $H = 1$ checks and sampling intervals are in Appendix C.4.4.

H	N	m	γ/γ_b	Cosine	Projection ratio	Relative residual
2	32	8	1	0.997719	0.938327	0.088505
4	8	32	1	0.996500	0.715730	0.290542
4	32	8	1/4	0.997252	0.935781	0.094644
4	32	8	1/2	0.998972	0.867499	0.138225
4	32	8	1	0.998994	0.784711	0.218151
4	128	2	1	0.999254	0.796703	0.205615

The local-step comparison further separates these two aspects of agreement: at the base step size and $N = 32, m = 8$, the observed $H = 4/H = 2$ projected ratio is 2.927, compared with the analytical ratio 3.5. Together, the results give a directional explanation of the finite-tail algorithm difference while identifying quantitative departures in magnitude and local-step scaling. Appendix C.4.4 gives the sampling intervals and window sensitivity.

7 Conclusion and future directions

Client averaging and control correction act on different parts of stationary mean bias. We made this distinction explicit for SCAFFOLD and matched FedAvg: the algorithms share a γ/N term, while their client-independent γ^2 coefficients separate ordinary local-update and persistent-control contributions. Local second moments coupled with third-order curvature explain the additional control effect, and a uniform remainder identifies the coefficients along arbitrary joint small-step, large-client limits. An exact control translation extends the SCAFFOLD result to affine objective shifts.

Controlled experiments examine these quantitative predictions. Real-data model averaging illustrates that finite-tail mean displacement can be an important component of parameter error, and a post-hoc full-vector comparison finds close directional agreement with the derived curvature–noise structure despite magnitude and local-step discrepancies. The analysis and experiments in this study explain an aspect of federated optimization that leading-order bias and variance alone do not distinguish.

Several directions follow from this coefficient-level characterization:

- **Nonlinear client heterogeneity:** Characterize how client-specific curvature and its interaction with local noise modify stationary bias beyond affine objective shifts.
- **Partial participation:** Determine how client subsampling and delayed control updates affect the stationary coefficients.

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Several directions follow from this coefficient-level characterization:

- **Nonlinear client heterogeneity:** Characterize how client-specific curvature and its interaction with local noise modify stationary bias beyond affine objective shifts.
- **Partial participation:** Determine how client subsampling and delayed control updates affect the stationary coefficients.
- **State-dependent noise:** Extend the analysis to gradient noise whose distribution varies with the local model.
- **Growing local-step counts:** Establish regimes in which H varies jointly with N and γ.
- **Bias-aware control design:** Investigate whether alternative control updates can mitigate second-order bias while accounting for drift correction, variance, and computational cost.

Code availability. The companion repository for the implementations, detailed configurations, numerical records, and reproduction instructions is https://github.com/Duck-YP/stationary-bias-study.

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Table 13: Sampling intervals and offsets for the fixed-step extrapolated intercepts in Figure 2.

j	95% simultaneous CI	Intercept − theory
1	$[-0.0489079, -0.0483504]$	-0.0006102
2	$[-0.0607622, -0.0600656]$	-0.0003902

Table 14: Five-dimensional SCAFFOLD coefficient targets, fitted intercepts, and nominal simultaneous 95% bootstrap intervals.

H	j	$\widehat{\beta}_{0,j}$	Theory	95% simultaneous CI
2	1	-0.0140655061	-0.0140249988	$[-0.0142709800, -0.0138295249]$
2	2	-0.0457104417	-0.0455796216	$[-0.0463785328, -0.0449434842]$
2	3	0.0322203570	0.0321284384	$[0.0316797243, 0.0326913914]$
2	4	-0.0340954937	-0.0339978616	$[-0.0345938212, -0.0335234201]$
2	5	0.0160014114	0.0159557997	$[0.0157329196, 0.0162353381]$
4	1	-0.0444005615	-0.0444124962	$[-0.0446094313, -0.0441636371]$
4	2	-0.1441927099	-0.1443354682	$[-0.1448724780, -0.1434224100]$
4	3	0.1016052163	0.1017400551	$[0.1010621370, 0.1020846930]$
4	4	-0.1075543334	-0.1076598952	$[-0.1080613670, -0.1069797680]$
4	5	0.0504598160	0.0505266990	$[0.0501901098, 0.0506979332]$

C.2.3 Fixed-step and five-dimensional results

Design and rationale. These studies separately examine client count and dimension using the objective family in Equation (74). The fixed-step study uses the grid in Table 7. The five-dimensional instance has positive-definite A, C with $\|AC - CA\|_F = 0.0876171216$.

Results and interpretation. Table 13 adds the fixed-step intercept intervals and offsets to Figure 2. Table 14 supplies all five coordinates for each local-step count. These details support the client-count and multidimensional interpretations in Sections 6.2.3 and 6.2.4.

C.2.4 Leading-coefficient and restricted-grid sensitivity

Design and rationale. The independent fits estimate the common leading coefficient directly. Restricted-grid comparisons omit the largest step size or, for the matched control fits, the smallest client count. This assesses sensitivity to points farther from the joint asymptotic limit.

Results and interpretation. Table 15 gives the individual leading-intercept estimates, complementing the matched results in Table 3. The $H = 8$ offsets persist as a sensitivity issue even when restricted fits move toward the targets. Primary estimates remain the basis of the comparisons in Section 6.2.5.

C.2.5 Contraction and noise-law resolution

Design and rationale. The supplementary comparisons assess how the matched result relates to individual coefficients, the second coordinate, and restricted grids. All use the model and resolution margins in Appendix C.1.4. The signed-covariance contrast subtracts the $\rho = -1/4$ symmetric condition from $\rho = 1/4$.

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H	j	$\widehat{\beta}_{0,j}$	Theory	95% simultaneous CI
2	1	-0.0140655061	-0.0140249988	$[-0.0142709800, -0.0138295249]$
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Table 15: Independent leading-coefficient fits with nominal 95% intervals. S: SCAFFOLD; F: FedAvg. The targets are $B_{11,1} = -0.02445943$ and $B_{11,2} = -0.03057429$ for every H; $H = 1$ checks the boundary recursion.

H	Alg.	Estimate_1	Interval_1	Estimate_2	Interval_2
1	S	-0.024531	$[-0.024960, -0.024102]$	-0.030664	$[-0.031200, -0.030127]$
1	F	-0.024531	$[-0.024960, -0.024102]$	-0.030664	$[-0.031200, -0.030127]$
2	S	-0.024432	$[-0.025068, -0.023795]$	-0.030554	$[-0.031349, -0.029758]$
2	F	-0.024413	$[-0.025051, -0.023774]$	-0.030525	$[-0.031323, -0.029728]$
4	S	-0.025169	$[-0.025953, -0.024385]$	-0.031537	$[-0.032517, -0.030557]$
4	F	-0.024799	$[-0.025574, -0.024024]$	-0.031044	$[-0.032013, -0.030075]$
8	S	-0.028252	$[-0.029046, -0.027459]$	-0.035610	$[-0.036602, -0.034618]$
8	F	-0.025913	$[-0.026712, -0.025114]$	-0.032560	$[-0.033559, -0.031562]$

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Design and rationale. The independent fits estimate the common leading coefficient directly. Restricted-grid comparisons omit the largest step size or, for the matched control fits, the smallest client count. This assesses sensitivity to points farther from the joint asymptotic limit.

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Design and rationale. The supplementary comparisons assess how the matched result relates to individual coefficients, the second coordinate, and restricted grids. All use the model and resolution margins in Appendix C.1.4. The signed-covariance contrast subtracts the $\rho = -1/4$ symmetric condition from $\rho = 1/4$.

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Results and interpretation. Table 16 supplements the matched first-coordinate results in Table 4. The paired signed-covariance estimates and simultaneous intervals are

$$\begin{aligned} j = 1: &\quad -0.04372973, &&[-0.04378778, -0.04367168], \\ j = 2: &\quad -0.00002676, &&[-0.00005541, 0.00000188], \end{aligned}$$

against target $(-0.04375, 0)^\top$. Individual first-coordinate null and noise-law intervals are wider than their resolution bands, so the precision gain is concentrated in the paired increment.

Restricted-grid sensitivity. Table 17 records the three restricted-grid comparisons whose simultaneous intervals narrowly exclude their targets. The two individual coefficient estimates lose sign resolution. All three corresponding Bonferroni-t intervals include the targets, showing sensitivity to the interval construction as well as the grid.

For the $\rho = 1/4$ symmetric paired second coordinate, the primary interval excludes zero but is contained in the resolution band. Its Bonferroni-t and restricted-grid intervals include zero. Reflection symmetry of this objective and symmetric-noise recursion fixes the ideal second-coordinate mean at zero from zero initial states. The small empirical offset therefore probes finite-simulation and interval accuracy, while the resolution-band comparison measures the size of that offset.

First/last-half balance comparisons give eight nominal 95% exclusions among 144 scalar diagnostics, all in individual algorithms; paired intervals contain zero. Direct means have substantially larger sampling variation than balance estimates. The diagnostics assess complementary sources of uncertainty: window comparisons assess temporal variation, finite-window identities assess arithmetic and alignment, and restricted fits assess grid sensitivity. Stationary approximation remains unresolved.

C.3 Estimator and structural consistency checks

C.3.1 Direct–balance estimator checks

Design and rationale. The direct global mean provides a cross-check of the primary nonlinear balance observable. The unnormalized diagnostic is

$$|\widehat{b}_{\mathrm{direct},j} - \widehat{b}_{\mathrm{bal},j}| \le 3\,\mathrm{SE}_{\mathrm{comb}}, \qquad \mathrm{SE}_{\mathrm{comb}} = \sqrt{\mathrm{SE}^2_{\mathrm{direct},j} + \mathrm{SE}^2_{\mathrm{bal},j}}.$$

Using the unnormalized means keeps the comparison on the parameter scale; second-order normalization multiplies estimation errors by γ^{-2}.

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against target $(-0.04375, 0)^\top$. Individual first-coordinate null and noise-law intervals are wider than their resolution bands, so the precision gain is concentrated in the paired increment.

Table 16: Additional contraction and noise-law results: individual coefficients and contrasts, and the paired second coordinate. Estimates and interval endpoints are multiplied by 10^3. S: SCAFFOLD; F: FedAvg; D: matched difference. Targets follow Equation (82). The law contrast is symmetric minus asymmetric at $\rho = 1/4$; the covariance contrast is $\rho = 1/4$ minus $-1/4$ with symmetric noise. Intervals use the primary family in Table 4.

Condition / contrast	Alg.	Estimate$_1$	Interval$_1$	Estimate$_2$	Interval$_2$
$-1/4$, sym.	S	56.0908	[42.4573, 69.7243]	2.6156	[−4.5559, 9.7871]
$-1/4$, sym.	F	34.2122	[20.5757, 47.8487]	2.6118	[−4.5609, 9.7844]
0, sym.	S	4.6861	[−18.0942, 27.4663]	2.4973	[−6.0117, 11.0064]
0, sym.	F	4.7066	[−18.0689, 27.4822]	2.5118	[−5.9994, 11.0230]
$+1/4$, sym.	S	−46.8325	[−67.9207, −25.7443]	2.2185	[−4.0669, 8.5038]
$+1/4$, sym.	F	−24.9813	[−46.0714, −3.8913]	2.2414	[−4.0463, 8.5291]
$+1/4$, asym.	S	−58.1461	[−81.4153, −34.8769]	2.7156	[−4.1149, 9.5460]
$+1/4$, asym.	F	−36.2643	[−59.5301, −12.9986]	2.7160	[−4.1137, 9.5457]
Law contrast	S	11.3136	[−20.1352, 42.7624]	−0.4971	[−9.8863, 8.8921]
Law contrast	F	11.2830	[−20.1627, 42.7287]	−0.4747	[−9.8649, 8.9156]
Covariance contrast	S	−102.9233	[−127.9666, −77.8799]	−0.3971	[−9.8315, 9.0372]
Covariance contrast	F	−59.1935	[−84.2420, −34.1451]	−0.3704	[−9.8073, 9.0665]
Additional paired second-coordinate results (zero target)					
$-1/4$, sym.	D	—		0.0038	[−0.0175, 0.0251]
0, sym.	D	—		−0.0145	[−0.0349, 0.0059]
$+1/4$, sym.	D	—		−0.0229	[−0.0423, −0.0035]
$+1/4$, asym.	D	—		−0.0005	[−0.0213, 0.0204]
Law contrast	D	—		−0.0224	[−0.0511, 0.0062]

Restricted-grid sensitivity. Table 17 records the three restricted-grid comparisons whose simultaneous intervals narrowly exclude their targets. The two individual coefficient estimates lose sign resolution. All three corresponding Bonferroni-t intervals include the targets, showing sensitivity to the interval construction as well as the grid.

For the $\rho = 1/4$ symmetric paired second coordinate, the primary interval excludes zero but is contained in the resolution band. Its Bonferroni-t and restricted-grid intervals include zero. Reflection symmetry of this objective and symmetric-noise recursion fixes the ideal second-coordinate mean at zero from zero initial states. The small empirical offset therefore probes finite-simulation and interval accuracy, while the resolution-band comparison measures the size of that offset.

First/last-half balance comparisons give eight nominal 95% exclusions among 144 scalar diagnostics, all in individual algorithms; paired intervals contain zero. Direct means have

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$+1/4$, asym.	F	−36.2643	[−59.5301, −12.9986]	2.7160	[−4.1137, 9.5457]
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Law contrast	F	11.2830	[−20.1627, 42.7287]	−0.4747	[−9.8649, 8.9156]
Covariance contrast	S	−102.9233	[−127.9666, −77.8799]	−0.3971	[−9.8315, 9.0372]
Covariance contrast	F	−59.1935	[−84.2420, −34.1451]	−0.3704	[−9.8073, 9.0665]
Additional paired second-coordinate results (zero target)					
$-1/4$, sym.	D	—		0.0038	[−0.0175, 0.0251]
0, sym.	D	—		−0.0145	[−0.0349, 0.0059]
$+1/4$, sym.	D	—		−0.0229	[−0.0423, −0.0035]
$+1/4$, asym.	D	—		−0.0005	[−0.0213, 0.0204]
Law contrast	D	—		−0.0224	[−0.0511, 0.0062]

Table 17: First-coordinate sensitivity after removing the largest step in the contraction study. The first two rows use $\rho = 1/4$ and symmetric noise; the last row is the signed-covariance contrast.

Quantity	Target	Estimate	Simultaneous interval
SCAFFOLD coefficient	−0.059375	−0.01675171	[−0.05907084, 0.02556741]
FedAvg coefficient	−0.0375	0.00510137	[−0.03721159, 0.04741432]
SCAFFOLD covariance contrast	−0.11875	−0.06840058	[−0.11872928, −0.01807187]

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SCAFFOLD covariance contrast	−0.11875	−0.06840058	[−0.11872928, −0.01807187]

Table 18: The four direct–balance discrepancies in the primary FedAvg study. The diagnostic uses the unnormalized difference $|\Delta\widehat{b}|$; the normalized statistics are also shown.

H	N	j	$\widehat{C}_{\text{bal}}$	$\widehat{C}_{\text{dir}}$	$\lvert\Delta\widehat{b}\rvert$	$3\,\text{SE}_{\text{comb}}$
2	16	2	−0.0408082	0.924598	0.00377112	0.00331567
2	48	1	−0.0323449	3.070686	0.00134680	0.00115744
4	16	2	−0.1080240	0.841309	0.00370833	0.00334677
4	48	1	−0.0933450	2.615233	0.00117560	0.00110953

substantially larger sampling variation than balance estimates. The diagnostics assess complementary sources of uncertainty: window comparisons assess temporal variation, finite-window identities assess arithmetic and alignment, and restricted fits assess grid sensitivity. Stationary approximation remains unresolved.

C.3 Estimator and structural consistency checks

C.3.1 Direct–balance estimator checks

Design and rationale. The direct global mean provides a cross-check of the primary nonlinear balance observable. The unnormalized diagnostic is

$$|\widehat{b}_{\text{direct},j} - \widehat{b}_{\text{bal},j}| \le 3\,\text{SE}_{\text{comb}}, \qquad \text{SE}_{\text{comb}} = \sqrt{\text{SE}^2_{\text{direct},j} + \text{SE}^2_{\text{bal},j}}.$$

Using the unnormalized means keeps the comparison on the parameter scale; second-order normalization multiplies estimation errors by γ^{-2}.

Results and interpretation. The primary two-dimensional FedAvg study satisfies the diagnostic in 156 of 160 chain-coordinate checks. Table 18 gives the four exceptions; their discrepancy-to-threshold ratios are 1.137, 1.164, 1.108, and 1.060. All checks satisfy the criterion in the two-dimensional SCAFFOLD studies, fixed-step study, five-dimensional SCAFFOLD study, and smaller-step FedAvg study. These counts concern the individual algorithms. In the independent controlled study, four of 68 algorithm–configuration raw-versus-balance intervals exclude zero. The primary estimates throughout use the balance observable.

Reanalysis of the existing $H = 4$ Cartesian-grid runs gives paired raw-minus-balance fitted-intercept differences of -2.39×10^{-7} and -2.92×10^{-7}. The paired intercepts therefore change little under estimator substitution. Finite-window identities agree where endpoints are available. Missing endpoints and individual noise sums limit this check to those windows; the joint-grid $H = 4$ discrepancy remains as reported in Appendix C.2.2.

C.3.2 Orthogonal coordinate equivariance

Design and rationale. A coordinate change should transform the complete recursion and its coefficient vectors consistently. For $y = Qx$ with $QQ^\top = I$, set $A_y = QAQ^\top$, $C_y = QCQ^\top$, and

Table 18: The four direct–balance discrepancies in the primary FedAvg study. The diagnostic uses the unnormalized difference $|\Delta\widehat{b}|$; the normalized statistics are also shown.

H	N	j	$\widehat{C}_{\mathrm{bal}}$	$\widehat{C}_{\mathrm{dir}}$	$\lvert\Delta\widehat{b}\rvert$	$3\,\mathrm{SE}_{\mathrm{comb}}$
2	16	2	−0.0408082	0.924598	0.00377112	0.00331567
2	48	1	−0.0323449	3.070686	0.00134680	0.00115744
4	16	2	−0.1080240	0.841309	0.00370833	0.00334677
4	48	1	−0.0933450	2.615233	0.00117560	0.00110953

Table 19: Structural consistency checks. Corresponding initial states and noise realizations are used within each comparison.

Check	Configuration
Orthogonal rotation	$Q = \left(\begin{smallmatrix} 0.8660254037844386 & -0.5 \\ 0.5 & 0.8660254037844386 \end{smallmatrix}\right)$ (30°); $H \in \{2, 4\}$; $(N, \gamma) \in \{(16, 1/16), (64, 1/64)\}$; 4 paired runs per configuration, 16 paths in total
Affine translation	$N \in \{8, 32\}$; $H \in \{2, 4, 8\}$; $a_c = \pm\rho u$, $\rho \in \{0, 0.25, 0.5, 1\}$; $\gamma = 1/128$; 24 configurations of 2,000 rounds

Results and interpretation. The primary two-dimensional FedAvg study satisfies the diagnostic in 156 of 160 chain-coordinate checks. Table 18 gives the four exceptions; their discrepancy-to-threshold ratios are 1.137, 1.164, 1.108, and 1.060. All checks satisfy the criterion in the two-dimensional SCAFFOLD studies, fixed-step study, five-dimensional SCAFFOLD study, and smaller-step FedAvg study. These counts concern the individual algorithms. In the independent controlled study, four of 68 algorithm–configuration raw-versus-balance intervals exclude zero. The primary estimates throughout use the balance observable.

Reanalysis of the existing $H = 4$ Cartesian-grid runs gives paired raw-minus-balance fitted-intercept differences of -2.39×10^{-7} and -2.92×10^{-7}. The paired intercepts therefore change little under estimator substitution. Finite-window identities agree where endpoints are available. Missing endpoints and individual noise sums limit this check to those windows; the joint-grid $H = 4$ discrepancy remains as reported in Appendix C.2.2.

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Design and rationale. A coordinate change should transform the complete recursion and its coefficient vectors consistently. For $y = Qx$ with $QQ^\top = I$, set $A_y = QAQ^\top$, $C_y = QCQ^\top$, and $S_{C,y} = QS_CQ^\top$. The Lyapunov equation becomes $A_yS_{C,y} + S_{C,y}A_y = C_y$, and

$$T_y[QMQ^\top] = QT[M] \quad \text{for every symmetric } M. \tag{83}$$

The coefficient formulas then give

$$B_{11,y} = QB_{11}, \qquad B^a_{20,y}(H) = QB^a_{20}(H), \quad a \in \{\mathrm{SCAFFOLD}, \mathrm{FedAvg}, \mathrm{ctrl}\}. \tag{84}$$

The simulation applies the same transformation to every noise draw. Table 19 lists the rotation and affine comparison settings.

Results and interpretation. The full-path residuals satisfy

$$\max_t \|x^{(y)}_t - Qx^{(x)}_t\|_\infty \le 1.1491 \times 10^{-14}, \qquad \max_{t,c} \|\xi^{(y)}_{t,c} - Q\xi^{(x)}_{t,c}\|_\infty \le 1.5765 \times 10^{-14},$$

Table 19: Structural consistency checks. Corresponding initial states and noise realizations are used within each comparison.

Check	Configuration
Orthogonal rotation	$Q = \left(\begin{smallmatrix} 0.8660254037844386 & -0.5 \\ 0.5 & 0.8660254037844386 \end{smallmatrix}\right)$ (30°); $H \in \{2, 4\}$; $(N, \gamma) \in \{(16, 1/16), (64, 1/64)\}$; 4 paired runs per configuration, 16 paths in total
Affine translation	$N \in \{8, 32\}$; $H \in \{2, 4, 8\}$; $a_c = \pm\rho u$, $\rho \in \{0, 0.25, 0.5, 1\}$; $\gamma = 1/128$; 24 configurations of 2,000 rounds

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with $\max_{t,h} \|\bar{q}^{(y)}_{t,h} - Q\bar{q}^{(x)}_{t,h}\|_\infty \le 5.4210 \times 10^{-17}$ and coefficient residuals below 6×10^{-16}. All are within the 10^{-10} tolerance, confirming coordinate consistency of the SCAFFOLD implementation.

C.3.3 Affine translation and update identities

Design and rationale. The affine comparisons use balanced gradient offsets and the translated initial controls prescribed by Proposition 4.4. With common noise, the transformed states should agree at every round. Zero initialization in both untransformed systems gives different transients because the initial states are no longer translated counterparts. Long-run $H = 1$ checks separately assess arithmetic accumulation.

Results and interpretation. The affine residual reported in Section 6.2.6 reproduces the exact translation to numerical precision. Short-run update-identity checks also pass. Over longer runs, 48 of 132 $H = 1$ arithmetic checks exceed 10^{-10}, with maximum vector discrepancy about 7.54×10^{-10}. The shared-objective subset has 24 exceedances among 60 checks and maximum coordinate discrepancy 4.33×10^{-10}. Two additional paths using a stable endpoint-increment calculation reduce vector discrepancies to about 10^{-12}. This improvement is documented for those two paths; the original full-grid outputs remain the reported results.

C.4 Real-data analyses and diagnostics

C.4.1 Objectives, partitions, and allocation estimands

Design and rationale. The shared-objective comparison isolates allocation effects, while disjoint partitions introduce heterogeneous local objectives. We use the a9a training and test

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C.4 Real-data analyses and diagnostics

C.4.1 Objectives, partitions, and allocation estimands

Design and rationale. The shared-objective comparison isolates allocation effects, while disjoint partitions introduce heterogeneous local objectives. We use the a9a training and test sets from the LIBSVM collection.[1] Table 20 collects preprocessing, training, and averaging settings. Each client minimizes

$$f_c(x) = \frac{1}{|D_c|} \sum_{i \in D_c} \{\log(1 + e^{z_i^\top x}) - y_i z_i^\top x\} + \frac{\lambda}{2}\|x\|^2. \tag{85}$$

Equal client weights preserve the global empirical objective. The shared-objective reference aggregate covariance has trace 0.001060 after rounding; disjoint partitions can change reference covariances and local objectives, including under IID assignment.

The reference Newton solve has gradient norm 1.253×10^{-16} and strong-convexity parameter-error bound 1.253×10^{-14}; a separately initialized solver differs by 8.43×10^{-15}. These floating-point residuals quantify numerical accuracy of the reference used to evaluate A_{ref}, T_{ref}, and Σ_{ref}. The projection direction in Equation (35) is fixed before training.

Squared mean displacement is estimated by $\widehat{U}_K$ in Equation (86) at the complete tail. Raw-checkpoint MSE comes from separately accumulated squared errors; its variance estimate subtracts the corresponding cross-run squared-mean estimate. Negative sampling-corrected values are retained. The nominal jackknife-t intervals and adjusted contrast families are described in Table 9.

Results and interpretation. Table 21 adds raw error and held-out loss results to the main allocation study, and supplies the disjoint-partition results. In the label-skew case at $N = 32, H = 4$, deterministic FedAvg converges to a point about 0.08365 from x_{ref}. The algorithm difference therefore includes a substantial drift-correction effect in addition to stochastic mean displacement.

[1] https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/binary.html

Table 20: Real-data settings. All training configurations use twelve independent SCAFFOLD–FedAvg pairs. Windows within a run share its trajectory. The two longer-run pilots are analyzed separately.

Component	Specification
Training data	32,512 of 32,561 rows selected by a fixed random permutation to allow equal disjoint partitions
Features and penalty	Each 123-feature row divided by $\max\{1, \|z_i\|\}$, then augmented with an intercept; $y_i \in \{0, 1\}$; all 124 parameters regularized with $\lambda = 0.01$
Client objectives	Shared: all clients use the same data; disjoint IID: random partition; label skew: two label-sorted shards per client. Disjoint regimes use one primary and two alternative partitions
Minibatches and base step	Sampling with replacement; $L_{\mathrm{sample}} = 0.51$; $\gamma_b = 0.1/L_{\mathrm{sample}} = 0.1960784314$
Fixed-budget allocations	$Nm = 256$; $(N, m) \in \{(8, 32), (32, 8), (128, 2)\}$; $H = 4$, $\gamma = \gamma_b$
Additional shared-objective runs	Three step-size/local-step variants (Table 5) and three $H = 1$ checks; nine shared-objective configurations in total
Averaging windows	$K \in \{1, 2, 4, 8, 16, 32, 64\}$ final batches per run; primary configuration $H = 4, N = 128, m = 2, \gamma_b$
Primary averaging horizon	1,280 rounds per batch; 6,400 burn-in rounds; longest window 81,920 rounds, ending at round 88,320; 90,439,680 example-gradient evaluations per algorithm
Longer-run pilots	Two runs; 25,600 burn-in rounds and 128 batches of 1,280 rounds; 189,440 rounds per run

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The pointwise projection half-width target 10^{-4} is unmet in 32 of 35 real-data configurations, including the main shared-objective allocations. The separately estimated allocation contrasts

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Additional shared-objective runs	Three step-size/local-step variants (Table 5) and three $H = 1$ checks; nine shared-objective configurations in total
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Primary averaging horizon	1,280 rounds per batch; 6,400 burn-in rounds; longest window 81,920 rounds, ending at round 88,320; 90,439,680 example-gradient evaluations per algorithm
Longer-run pilots	Two runs; 25,600 burn-in rounds and 128 batches of 1,280 rounds; 189,440 rounds per run

The pointwise projection half-width target 10^{-4} is unmet in 32 of 35 real-data configurations, including the main shared-objective allocations. The separately estimated allocation contrasts nevertheless resolve their signed differences. Precision of an individual mean and precision of a paired contrast are distinct features of the experiment.

C.4.2 Run-length sensitivity

Design and rationale. First/last-half comparisons assess variation within the recorded tails. They use the same 64-batch trajectories and the 2×69 adjustment across controlled and real-data projected contrasts. In the shared-objective real-data study, the primary observable is the direct mean itself.

Results and interpretation. The adjusted first/last-half contrast intervals contain zero. The endpoint allocation contrasts in the first and last halves are 0.003098 and 0.003129 for SCAFFOLD, and 0.002053 and 0.002092 for FedAvg. The similar values support stability of the allocation comparison within the sampled tails. The stronger temporal-equivalence criterion is assessed separately below.

C.4.3 Finite-tail model averaging

Design and rationale. To quantify the components in Equation (36), set $e^{\mathcal{A}}_{r,K} = Y^{\mathcal{A}}_{r,K} - x_{\text{ref}}$ and $\bar{e}_K = R^{-1} \sum_r e_{r,K}$ with $R = 12$. Suppressing the algorithm superscript,

$$\widehat{U}_K = \frac{\sum_{r \neq s} e_{r,K}^\top e_{s,K}}{R(R-1)}, \qquad \widehat{V}_K = \frac{\sum_r \|e_{r,K} - \bar{e}_K\|^2}{R-1}, \qquad \widehat{M}_K = \frac{1}{R} \sum_r \|e_{r,K}\|^2 = \widehat{U}_K + \widehat{V}_K. \quad (86)$$

Cross-products between independent runs estimate the squared mean displacement, while between-run deviations estimate the variance of the time-averaged outputs. The estimators

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Table 21: Additional finite-tail allocation results at $H = 4$, γ_b, and $Nm = 256$. The upper panel supplements Figure 3 with error and loss measurements; the lower panel reports disjoint partitions. Projection intervals are nominal 95% intervals. Raw-checkpoint MSE is multiplied by 10^3; held-out loss is averaged over the per-run tail-average models. S: SCAFFOLD; F: FedAvg.

Shared objective: additional error and loss measurements				
N	MSE (S)	MSE (F)	Test loss (S)	Test loss (F)
8	2.3511	2.3509	0.424981	0.424983
32	2.3629	2.3618	0.424961	0.424970
128	2.3745	2.3655	0.424889	0.424922

N	Alg.	Projection	Nominal interval	MSE	Test loss
Disjoint IID partition					
8	S	0.000341	$[0.000213, 0.000468]$	2.3458	0.424978
8	F	0.000268	$[0.000140, 0.000396]$	2.3454	0.424981
32	S	0.000880	$[0.000638, 0.001123]$	2.3661	0.424963
32	F	0.000472	$[0.000230, 0.000715]$	2.3643	0.424977
128	S	0.003553	$[0.003393, 0.003714]$	2.3642	0.424884
128	F	0.001994	$[0.001830, 0.002158]$	2.3507	0.424936
Disjoint label-skew partition					
8	S	0.000243	$[0.000050, 0.000436]$	2.2526	0.424984
8	F	−0.039298	$[-0.039491, -0.039104]$	5.6791	0.426589
32	S	0.000352	$[0.000143, 0.000561]$	2.2180	0.424981
32	F	−0.055891	$[-0.056096, -0.055686]$	9.0666	0.427274
128	S	0.001256	$[0.001096, 0.001416]$	2.1956	0.424956
128	F	−0.062027	$[-0.062182, -0.061872]$	10.7951	0.427548

nevertheless resolve their signed differences. Precision of an individual mean and precision of a paired contrast are distinct features of the experiment.

C.4.2 Run-length sensitivity

Design and rationale. First/last-half comparisons assess variation within the recorded tails. They use the same 64-batch trajectories and the 2×69 adjustment across controlled and real-data projected contrasts. In the shared-objective real-data study, the primary observable is the direct mean itself.

Results and interpretation. The adjusted first/last-half contrast intervals contain zero. The endpoint allocation contrasts in the first and last halves are 0.003098 and 0.003129 for SCAFFOLD, and 0.002053 and 0.002092 for FedAvg. The similar values support stability of the allocation comparison within the sampled tails. The stronger temporal-equivalence criterion is assessed separately below.

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$$\widehat{U}_K = \frac{\sum_{r \neq s} e_{r,K}^\top e_{s,K}}{R(R-1)}, \qquad \widehat{V}_K = \frac{\sum_r \|e_{r,K} - \bar{e}_K\|^2}{R-1}, \qquad \widehat{M}_K = \frac{1}{R} \sum_r \|e_{r,K}\|^2 = \widehat{U}_K + \widehat{V}_K. \quad (86)$$

Table 21: Additional finite-tail allocation results at $H = 4$, γ_b, and $Nm = 256$. The upper panel supplements Figure 3 with error and loss measurements; the lower panel reports disjoint partitions. Projection intervals are nominal 95% intervals. Raw-checkpoint MSE is multiplied by 10^3; held-out loss is averaged over the per-run tail-average models. S: SCAFFOLD; F: FedAvg.

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N	MSE (S)	MSE (F)	Test loss (S)	Test loss (F)
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32	2.3629	2.3618	0.424961	0.424970
128	2.3745	2.3655	0.424889	0.424922

N	Alg.	Projection	Nominal interval	MSE	Test loss
Disjoint IID partition					
8	S	0.000341	$[0.000213, 0.000468]$	2.3458	0.424978
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32	F	0.000472	$[0.000230, 0.000715]$	2.3643	0.424977
128	S	0.003553	$[0.003393, 0.003714]$	2.3642	0.424884
128	F	0.001994	$[0.001830, 0.002158]$	2.3507	0.424936
Disjoint label-skew partition					
8	S	0.000243	$[0.000050, 0.000436]$	2.2526	0.424984
8	F	-0.039298	$[-0.039491, -0.039104]$	5.6791	0.426589
32	S	0.000352	$[0.000143, 0.000561]$	2.2180	0.424981
32	F	-0.055891	$[-0.056096, -0.055686]$	9.0666	0.427274
128	S	0.001256	$[0.001096, 0.001416]$	2.1956	0.424956
128	F	-0.062027	$[-0.062182, -0.061872]$	10.7951	0.427548

Table 22: Additional longest-window model-averaging statistics at $K = 64$, with approximate simultaneous 95% primary-family intervals. Parameter errors and training gaps are multiplied by 10^6; test loss is unscaled. The component shares and total-MSE contrast are reported in Section 6.4.

Quantity	SCAFFOLD	FedAvg
$\widehat{M}_K$	17.267 $[15.383, 19.151]$	10.878 $[9.459, 12.297]$
Train objective gap	0.725 $[0.630, 0.819]$	0.361 $[0.295, 0.426]$
Held-out log loss	0.424889 $[0.424879, 0.424899]$	0.424922 $[0.424912, 0.424932]$
Paired difference	Estimate [simultaneous interval]	
$\Delta\widehat{U}_K$	6.378 $[5.848, 6.908]$	
$\Delta\widehat{V}_K$	0.011 $[-0.036, 0.058]$	
Additional intervals for the loss contrasts reported in the main text:		
Δ train gap	$[0.336, 0.393]$	
Δ held-out log loss	$[-0.00003308, -0.00003231]$	

Cross-products between independent runs estimate the squared mean displacement, while between-run deviations estimate the variance of the time-averaged outputs. The estimators retain negative values of $\widehat{U}_K$. Training and test losses are evaluated on each run's own $Y_{r,K}$ and then averaged across runs.

The primary, secondary, and additional configurations use the schedules in Table 20. The $N = 8, 32$ allocations are secondary; six further configurations examine step size, local steps, and $H = 1$. Bootstrap resampling uses one common run-index vector across algorithms, configurations, and overlapping windows. Separate uncertainty families are listed in Table 9; all condition on the fixed dataset and recorded finite-tail protocol.

Results and interpretation. Table 22 gives absolute errors and losses complementing the main-text error shares. For $b_S = b_F + \delta$, the difference in squared means obeys $\|b_S\|^2 - \|b_F\|^2 = 2b_F^\top\delta + \|\delta\|^2$. The sampling-corrected decomposition is $\Delta\widehat{U} = 2(2.57965 \times 10^{-6}) + 1.21856 \times 10^{-6}$, showing the contribution of both the cross term and the squared algorithmic displacement. The total-MSE contrast corresponds to a 58.7% relative difference. For scale, $f(x_{\text{ref}}) = 0.4758413$ and the held-out contrast has magnitude about 0.00769% of reference test loss.

At $N = 32$ and 8, the longest-window SCAFFOLD/FedAvg mean-displacement shares are 14.97%/7.77% and 1.61%/0.97%, respectively. Their squared-mean intervals give weaker evidence against zero than the primary allocation. Together with the main result, these comparisons show that the error share depends on allocation as well as averaging length.

Temporal diagnostics. The predeclared equivalence test requires both a 10^{-3} vector-norm margin and a 10^{-4} projected-drift margin. None of 144 checks meets both requirements. In the primary SCAFFOLD comparison, the first/last 32-batch drift norm is 0.0013633, with upper bound 0.0054372 and projected interval $[-0.0005438, 0.0007484]$. The longest window inherits the 32-batch check because a separate equal-length 64-batch anchor is unavailable. Thus temporal equivalence remains unresolved.

For SCAFFOLD, $\widehat{U}_K \geq \widehat{V}_K$ at $K = 32, 64$, with a positive lower bound for $U - V$ under both interval procedures only at $K = 64$. The prespecified crossover requirement—two consecutive supported windows together with temporal equivalence—is unmet. FedAvg has no point-estimate crossover on this grid. The protocol's 10% parameter-MSE-share threshold is a researcher-chosen reference. The two longer-run pilots in Table 20 leave these temporal questions unresolved. The established result is the finite-tail error decomposition reported in Section 6.4.

Table 22: Additional longest-window model-averaging statistics at $K = 64$, with approximate simultaneous 95% primary-family intervals. Parameter errors and training gaps are multiplied by 10^6; test loss is unscaled. The component shares and total-MSE contrast are reported in Section 6.4.

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Held-out log loss	0.424889 $[0.424879, 0.424899]$	0.424922 $[0.424912, 0.424932]$
Paired difference	Estimate [simultaneous interval]	
$\Delta\widehat{U}_K$	6.378 $[5.848, 6.908]$	
$\Delta\widehat{V}_K$	0.011 $[-0.036, 0.058]$	
Additional intervals for the loss contrasts reported in the main text:		
Δ train gap	$[0.336, 0.393]$	
Δ held-out log loss	$[-0.00003308, -0.00003231]$	

retain negative values of $\widehat{U}_K$. Training and test losses are evaluated on each run's own $Y_{r,K}$ and then averaged across runs.

The primary, secondary, and additional configurations use the schedules in Table 20. The $N = 8, 32$ allocations are secondary; six further configurations examine step size, local steps, and $H = 1$. Bootstrap resampling uses one common run-index vector across algorithms, configurations, and overlapping windows. Separate uncertainty families are listed in Table 9; all condition on the fixed dataset and recorded finite-tail protocol.

Results and interpretation. Table 22 gives absolute errors and losses complementing the main-text error shares. For $b_S = b_F + \delta$, the difference in squared means obeys $\|b_S\|^2 - \|b_F\|^2 = 2b_F^\top\delta + \|\delta\|^2$. The sampling-corrected decomposition is $\Delta\widehat{U} = 2(2.57965 \times 10^{-6}) + 1.21856 \times 10^{-6}$, showing the contribution of both the cross term and the squared algorithmic displacement. The total-MSE contrast corresponds to a 58.7% relative difference. For scale, $f(x_{\text{ref}}) = 0.4758413$ and the held-out contrast has magnitude about 0.00769% of reference test loss.

At $N = 32$ and 8, the longest-window SCAFFOLD/FedAvg mean-displacement shares are 14.97%/7.77% and 1.61%/0.97%, respectively. Their squared-mean intervals give weaker evidence against zero than the primary allocation. Together with the main result, these comparisons show that the error share depends on allocation as well as averaging length.

Temporal diagnostics. The predeclared equivalence test requires both a 10^{-3} vector-norm margin and a 10^{-4} projected-drift margin. None of 144 checks meets both requirements. In the primary SCAFFOLD comparison, the first/last 32-batch drift norm is 0.0013633, with upper bound 0.0054372 and projected interval $[-0.0005438, 0.0007484]$. The longest window inherits the 32-batch check because a separate equal-length 64-batch anchor is unavailable. Thus temporal equivalence remains unresolved.

For SCAFFOLD, $\widehat{U}_K \geq \widehat{V}_K$ at $K = 32, 64$, with a positive lower bound for $U - V$ under both interval procedures only at $K = 64$. The prespecified crossover requirement—two consecutive supported windows together with temporal equivalence—is unmet. FedAvg has no point-estimate crossover on this grid. The protocol's 10% parameter-MSE-share threshold is a researcher-chosen reference. The two longer-run pilots in Table 20 leave these temporal questions unresolved. The established result is the finite-tail error decomposition reported in Section 6.4.

C.4.4 Post-hoc directional comparison

Design and rationale. To contextualize the alignment in Table 6, we compare the analytical direction with the fixed reference-parameter direction and the unpreconditioned vector $-T_{\text{ref}}[\Sigma_{\text{ref}}]$. Window, step-size, and local-step comparisons assess how alignment and amplitude change within the existing runs. The zero-vector proxy supplies an amplitude baseline. Sampling procedures are listed in Table 9; scalar and vector families each have their own 95% calibration.

Results and interpretation. At $K = 64$, reference-parameter cosines are 0.8953–0.9087 and unpreconditioned-direction cosines are 0.6590–0.6929. Both are lower than the analytical proxy values in the main table; the zero-vector proxy also has larger residuals in all six configurations. These comparisons make the geometric agreement specific to the Hessian-preconditioned curvature–noise direction among the tested references.

For the fixed-H, N, m sequence, adjacent log slopes of the observed projection are 1.8553 and 1.8907. Across shorter averaging windows, the cosine falls as low as 0.872197. These results describe finite-grid and window sensitivity. Allocation comparisons change N and m together, preserving total sampling budget while changing local covariance.

For the $H = 4/H = 2$ ratio in Section 6.5, the exploratory simultaneous interval is $[2.7415, 3.1125]$, the jackknife interval is $[2.7101, 3.1439]$, and the projection-band transform is $[2.6928, 3.1919]$. All exclude 3.5 and include the ordinary local-update ratio 3. The results resolve a departure from the plug-in local-step scaling; attribution of that departure remains open. The $H = 1$ residuals are approximately 10^{-11}–10^{-10}, consistent with the expected zero-proxy numerical identity.

The 54 early/late paired contrasts assess temporal variation of the algorithm difference. The model-averaging diagnostics in Appendix C.4.3 assess each algorithm separately, so the two comparisons address different temporal quantities. The proxy agreement remains a descriptive geometric result for the recorded finite-tail means.

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